



Facilitator Guide

Sector
Green Jobs

Sub Sector
Bio-Energy

Occupation

Operation and Maintenance

Reference Id

SGJ/MCr0017, Version 1.0; NSQF Level 4



Essentials of Biogas Plant Operations



Please Scan/ Click QR Code to access e book

Phone: 011-41792866

Disclaimer

SCGJ is not responsible for the views expressed or content or reliability of linked videos. SCGJ cannot guarantee that these links/QR codes will work all the time as we have no control over availability of linked pages.





“

Skilling is building a better India.
If we have to move India towards
development then Skill Development
should be our mission.

”

Shri Narendra Modi
Prime Minister of India





Certificate

COMPLIANCE TO QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS

is hereby issued by the

SKILL COUNCIL FOR GREEN JOBS

for the

SKILLING CONTENT: FACILITATOR GUIDE

Complying to National Occupational Standards of
Job Role/ Qualification Pack: 'Essentials of Biogas Plant Operations' QP No. 'SGJ/MC0017NSQF Level 4'

Date of Issuance:

Feb 13th, 2025

Valid up to next review date of Qualification: Feb 12th, 2029

Authorised Signatory
(Skill Council for Green Jobs)

Acknowledgements

This Facilitator Guide for the **Essentials of Biogas Plant Operations** qualification is a sincere effort to provide comprehensive and relevant information for aspiring and existing professionals. The content has been developed with inputs from Subject Matter Experts (SMEs), industry practitioners, field-level operators, and the support of **Greenloop Clean Technologies**, ensuring technical accuracy and practical relevance aligned with current green energy practices.

We extend our sincere gratitude to all contributors, including Greenloop Clean Technologies, for their valuable expertise and support in developing this Guide. We also acknowledge the support of the **Skill Council for Green Jobs (SCGJ)**, National Skill Development Corporation (NSDC), and the Ministry of Skill Development and Entrepreneurship, Government of India.

This handbook aims to support effective skill-based training and benefit all stakeholders working towards a sustainable future.



About this Book

The Facilitator Guide for "**Essentials of Biogas Plant Operations**" is developed to equip learners with the knowledge and practical skills required to operate and manage biogas plants efficiently. It provides a balanced understanding of both technical processes and on-ground operational practices within the biogas ecosystem.

The Guide introduces learners to the fundamentals of biogas technology and highlights its importance in sustainable waste management and renewable energy generation. It explains the key components of a biogas plant and guides participants in feedstock selection, preparation, and maintaining optimal conditions for anaerobic digestion. Learners are also trained to monitor plant performance, maintain records, and ensure smooth day-to-day operations.

The content further focuses on troubleshooting common operational issues, preventive maintenance practices, and ensuring plant efficiency. It also covers safety measures and protocols essential for working in biogas plant environments. Real-world insights and practical approaches are included to enhance understanding and applicability in field conditions.

Additionally, this Guide integrates essential employability skills such as communication, teamwork, and workplace safety awareness, ensuring that participants are not only technically skilled but also professionally prepared to contribute effectively to the green energy sector.

Table of Contents

1. Biogas Technology – Foundations and Understanding for Beginners	1
Unit 1.1 Introductory Overview	3
UNIT 1.2: Biochemical Pathways and Stages of Anaerobic Digestion	5
UNIT 1.3: Parameters Affecting Biogas Production	6
UNIT 1.4: Composition, Properties, and Energy Value of Biogas.....	7
UNIT 1.5: Advantages of Biogas Over Traditional Fuels.....	8
UNIT 1.6: Cleaning of Biogas	9
UNIT 1.7: Upgrading of Biogas	10
UNIT 1.8: Applications of Biogas and CBG	11
2. Feedstock Management and Pre-treatment	13
UNIT 2.1: Feedstock Characteristics and Digester	15
UNIT 2.2: Feedstock Classification for Biogas Production	16
UNIT 2.3: Key Feedstock Parameters Affecting Anaerobic.....	17
UNIT 2.4: Feedstock Management Practices.....	18
Unit 2.5 Feedstock Storage Management	20
UNIT 2.6: Pre-treatment for Feedstock	22
UNIT 2.7: Operational Considerations and Safety while Pre-Treatment of Feedstock	23
Unit 2.8 Health, Hygiene, and Safety Practices.....	24
3. Biogas Plant Components, Layout, and ancillary system.....	26
Unit 3.1 Introduction to Industrial Fixed-Dome Biogas Plants	28
UNIT 3.2: Introduction to Floating-Dome Biogas Digesters.....	32
UNIT 3.3: Introduction to CSTR Systems.....	36
UNIT 3.4: Introduction to Plug-Flow Biogas Plants.....	40
UNIT 3.5: Introduction to Ancillary Systems in Biogas and CBG Plants	46
UNIT 3.6: Overview of CHP Systems in Biogas and CBG Plants	51
4. Monitoring of Biogas Plants	56
UNIT 4.1: Introduction to Monitoring.....	58
UNIT 4.2: Standard Operating Procedures (SOPs) for Biogas Production.....	59
UNIT 4.3: Sampling Protocols.....	61
Unit - 4.4 Digestate Management & Utilization	62

UNIT 4.5: Advanced Monitoring and Laboratory Parameters.....	66
UNIT 4.6: Compliance, Environmental Monitoring, Preventive & Corrective Maintenance.....	67
UNIT 4.7: Troubleshooting, Safety, Compliance & Emergency Protocols.....	70
UNIT 4.8: Seasonal O&M Adjustments.....	71
UNIT 4.9: Troubleshooting SOPs, Safety, PPE & Emergency Protocols.....	72
 5. Health, Hygiene, and Safety Practices in Biogas Plants	74
UNIT 5.1: Introduction to Health and Hygiene Fundamentals in Biogas Environments.....	76
UNIT 5.2: Importance of Hygiene in Biogas Operations.....	79
UNIT 5.3: Operator Hygiene and Personal Cleanliness.....	83
UNIT 5.4: Hygiene Infrastructure and Regulatory Compliance.....	88
UNIT 5.5: Hazard Identification in Biogas Plants.....	90
Unit 5.6 Personal Protective Equipment (PPE).....	91
Unit 5.7 Safe Work Area and Housekeeping Practices.....	96
Unit 5.8 Gas Leak Detection.....	99
Unit 5.9 Fire Safety and Extinguisher Use.....	101
 6. SCADA Monitoring, Interpretation & Decision-Making in Biogas Plant O&M.....	108
UNIT 6.1 – Overview of SCADA in Biogas and CBG Plant Operations.....	110
UNIT 6.2: SCADA Architecture & Components.....	111
UNIT 6.3: Real-Time Monitoring Capabilities.....	112
Unit 6.4 SCADA Alarms & Alert Management	113
Unit 6.5 Trend Interpretation & Data Analytics	114
Unit 6.6-Tiered SCADA Decision-Making Framework.....	115
Unit 6.7- SCADA for Predictive & Preventive Maintenance.....	116
 Annexure	11818





1. Biogas Technology – Foundations and Understanding for Beginners

Unit 1.1 - Introduction

Unit 1.2 – Biochemical Pathways and Stages of Anaerobic Digestion

Unit 1.3 - Parameters Affecting Biogas Production

Unit 1.4 - Composition, Properties, and Energy Value of Biogas

Unit 1.5 - Advantages of Biogas Over Traditional Fuels

Unit 1.6 - Cleaning of Biogas

Unit 1.7 - Upgrading of Biogas

Unit 1.8 - Applications of Biogas and CBG



Key Learning Outcomes

At the end of this module, you will be able to:

1. Understand what biogas is and how it is produced from organic waste.
2. Explain the basic stages of digestion and how microorganisms produce gas.
3. Identify factors that affect gas production, such as feed, temperature, and PH.
4. Follow correct feeding and operating practices to keep digestion stable.
5. Understand why biogas must be cleaned and upgraded before use.
6. Recognise where biogas and CBG are used in daily life and industry.

Unit 1.1 Introductory Overview

Unit Objectives

At the end of this unit, you will be able to:

1. Explain how microorganisms convert organic waste into biogas and digestate
2. Recognise the role of the digester in linking waste management, energy, and agriculture

Ask

- Ask the participants what they think happens inside a biogas digester after waste is fed.
- Ask the participants if they believe gas production depends more on the tank structure or on what happens inside the slurry.

Notes for Facilitation

- Most learners initially view a biogas plant as a civil or mechanical structure rather than a biological system.
- Allow participants to describe the plant in their own words before guiding them.

Explain

- Explain that a biogas plant works because microorganisms slowly break down organic waste in the absence of oxygen.
- Explain that the digester provides a safe and controlled environment where these microorganisms can work continuously.



Figure:1- Biogas Plant

UNIT 1.2: Biochemical Pathways and Stages of Anaerobic Digestion

Unit Objectives

At the end of this unit, you will be able to:

1. Explain that biogas production happens through multiple biological stages
2. Identify the four main stages of anaerobic digestion

Ask

- Ask the participants if they think all microorganisms inside the digester do the same work.
- Ask them whether methane is produced throughout digestion or only at the end.

Notes for Facilitation

- Learners often assume digestion is a single step.

Explain

- Explain that biogas generation happens through multiple oxygen-free biological stages, each carried out by different groups of microorganisms.
- Explain that these microorganisms work in sequence and depend on each other.
- Explain that methane is mainly produced in the final stage, not at the beginning of digestion.

UNIT 1.3: Parameters Affecting Biogas Production

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the three main groups of parameters affecting biogas production
2. Explain how feedstock quality influences digestion behaviour

Ask

- Ask the participants what usually changes first when gas production starts dropping.
- Ask whether gas problems always appear suddenly or develop slowly over time.

Notes for Facilitation

- This is a core O&M section and should not be rushed.
- Participants often recognise symptoms but do not connect them to underlying parameters.

Explain

- Explain that biogas production depends on multiple factors working together inside the digester.

UNIT 1.4: Composition, Properties, and Energy Value of Biogas

Unit Objectives

At the end of this unit, you will be able to:

1. Explain why biogas quality reflects digestion conditions inside the digester
2. Identify the main components of biogas and their effects on safety and performance
3. Understand key physical properties of biogas relevant to operation

Ask

- Ask the participants what changes they notice first when gas quality becomes poor.
- Ask whether the amount of gas and the strength of gas are always the same.
- Ask the participants if they know they can tell

Notes for Facilitation

- Participants often focus only on gas quantity and ignore quality.
- Encourage them to recall experiences with weak flames, yellow flames, corrosion, or blocked pipelines.
- Keep the explanation practical and observation based.

Team Activity

- Observe and compare different biogas flame conditions.
- Match biogas components with their effects on safety and performance.
- Identify common gas quality problems and suggest solutions.

UNIT 1.5: Advantages of Biogas Over Traditional Fuels

Unit Objectives

At the end of this unit, you will be able to:

1. Explain why biogas is a reliable and renewable energy source
2. Compare biogas with traditional fuels such as LPG, diesel, and firewood
3. Understand how gas energy value reflects digestion health

Ask

- Ask the participants what they do when the biogas flame becomes weak.
- Ask whether fuel availability or fuel price has ever affected their daily work or household activities.

Notes for Facilitation

- Participants may think of advantages only in terms of cost savings.
- Link advantages back to daily operation and observation skills.
Avoid presenting this as promotional content. Keep it operational and experience-based.

Explain

- Explain that the strength of the biogas flame reflects the energy value of the gas.
- Explain that a weak or unstable flame usually indicates changes in digestion conditions such as acidity, temperature variation, or irregular feeding.

UNIT 1.6: Cleaning of Biogas

Unit Objectives

At the end of this unit, you will be able to:

1. Explain why raw biogas must be cleaned before use

Ask

- Ask the participants if they have ever noticed a yellow or weak flame even when gas pressure looked normal.
- Ask whether they have experienced frequent burner choking, corrosion, or unusual smell near gas pipelines.

Notes for Facilitation

- Operators often realise the need for cleaning only after damage has already started.
- Use this section to shift thinking from corrective repair to preventive protection.

Explain

- Explain that biogas produced inside the digester is rich in energy but contains impurities formed naturally during digestion.
- Explain that these impurities behave differently once gas leaves the warm, wet environment of the digester.

UNIT 1.7: Upgrading of Biogas

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain what biogas upgrading means and why it is required
- 2 Understand how carbon dioxide affects gas energy value and end use

Ask

- Ask the participants if they think gas used for cooking and gas used for vehicles are the same.
- Ask whether higher gas pressure alone makes gas suitable for engines or transport.

Notes for Facilitation

- Avoid deep chemical theory. Focus on function, outcome, and operator relevance. Link upgrading to discipline in digestion and cleaning, not just equipment installation.

Explain

- Explain that these impurities behave differently once gas leaves the warm, wet environment of the digester.

UNIT 1.8: Applications of Biogas and CBG

Unit Objectives

At the end of this unit, you will be able to:

- 1 Identify different applications of biogas based on gas quality
- 2 Distinguish between uses of raw biogas, cleaned biogas, and CBG

Ask

- Ask the participants where biogas is currently being used in their plant or community.
- Ask whether the same gas can be used for cooking, engines, and vehicles without any change.

Notes for Facilitation

- Participants often associate biogas only with cooking. Use this section to expand their understanding from household use to commercial and transport applications.

Explain

- Explain –Household and Community Uses
- Explain that biogas can be used directly in homes and communities when handled safely.
- Explain that it reduces dependence on LPG and firewood.



Key Learning Outcomes

At the end of this module, you will be able to:

- 1 Interpret dome movement and slurry behaviour as indicators of gas production and plant health.
- 2 Carry out routine operation and maintenance activities for stable and efficient performance.
- 3 Recognise common operational problems and apply basic troubleshooting measures.
- 4 Follow safety practices related to gas handling, leak detection, and daily plant operation.

UNIT 2.1: Feedstock Characteristics and Digester

Unit Objectives

At the end of this unit, you will be able to:

- 1 Understand how feedstock quality directly affects digestion stability and gas yield
- 2 Identify different types of organic materials used as biogas feedstock

Ask

- Ask the participants what materials are commonly fed into their digester.
- Ask whether gas problems they have seen usually started after a change in feeding material or pattern.

Notes for Facilitation

- Many operators see feedstock as waste rather than as a controlled input.
- Use this section to change that mindset from disposal to process control.

Explain

- Explain that feedstock is the main input material that microorganisms convert into biogas.
- Explain that once feedstock enters the digester in an oxygen-free environment, bacteria slowly break it down into methane, carbon dioxide, and small amounts of other gases.
- Explain that how well the digester performs depends directly on what feedstock is used and how it is prepared before feeding.

UNIT 2.2: Feedstock Classification for Biogas Production

Unit Objectives

At the end of this unit, you will be able to:

- 1 Classify feedstocks based on degradation speed, nutrient profile, and structure
- 2 Use feedstock classification to plan blending, pre-treatment, and feeding strategy

Ask

- Ask whether all organic waste can be fed in the same way and quantity.
- Ask the participants if they know they can tell

Explain

- Explain that feedstock entering a biogas plant comes from many sources such as farms, kitchens, markets, agro-industries, and crop fields.
- Explain that each of these materials behaves differently inside the digester because their chemical composition, moisture content, nutrient balance, particle size, and structural complexity vary widely.

Team Activity

- Classify different feedstocks based on degradation speed and structure.
- Compare various organic wastes and discuss suitable feeding methods.
- Plan a simple feedstock blending and pre-treatment strategy.

UNIT 2.3: Key Feedstock Parameters Affecting Anaerobic

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain how feedstock parameters control digestion stability and methane yield
- 2 Understand the role of C:N ratio, pH, moisture, OLR, and alkalinity

Ask

- Ask the participants which single change in feeding has caused the biggest gas fluctuation in their plant.
- Ask whether increasing feed quantity always increases gas production.

Notes for Facilitation

- This section translates numbers into operator decisions.
- Do not rush through ranges. Explain what happens when limits are crossed.
- Encourage participants to relate values to what they see, smell, and feel in the plant.

Explain

- Explain that the biological stability of a digester depends on maintaining the right feedstock parameters.
- Explain that these parameters define the environment in which microorganisms operate and directly control digestion rate, methane yield, and efficiency.

Table 1: C/N Ratio

Condition	Process Effect	Corrective Action
High C:N	Slow digestion, low yield	Add nitrogen-rich feed (manure, poultry litter)
Low C:N	Ammonia toxicity, high pH	Add carbon-rich feed (straw, fruit waste)

UNIT 2.4: Feedstock Management Practices

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain why feedstock management starts before material enters the digester

Ask

- Ask the participants what usually happens to feedstock immediately after it enters the plant.
- Ask whether they have seen gas problems that started after poor-quality or mixed waste was fed.

Notes for Facilitation

- This section is about discipline, not machinery.
- Emphasise that feedstock handling is a process control activity, not housekeeping.

Team Activity

- Discuss common problems caused by poor feedstock handling.
- Identify correct and incorrect feedstock management practices.
- Prepare a simple feedstock handling and monitoring plan.

Explain

- Explain that effective feedstock management begins the moment organic material enters the biogas plant.
- Explain that receiving, inspection, weighing, segregation, labelling, and storage determine whether digestion will remain stable or become unpredictable.

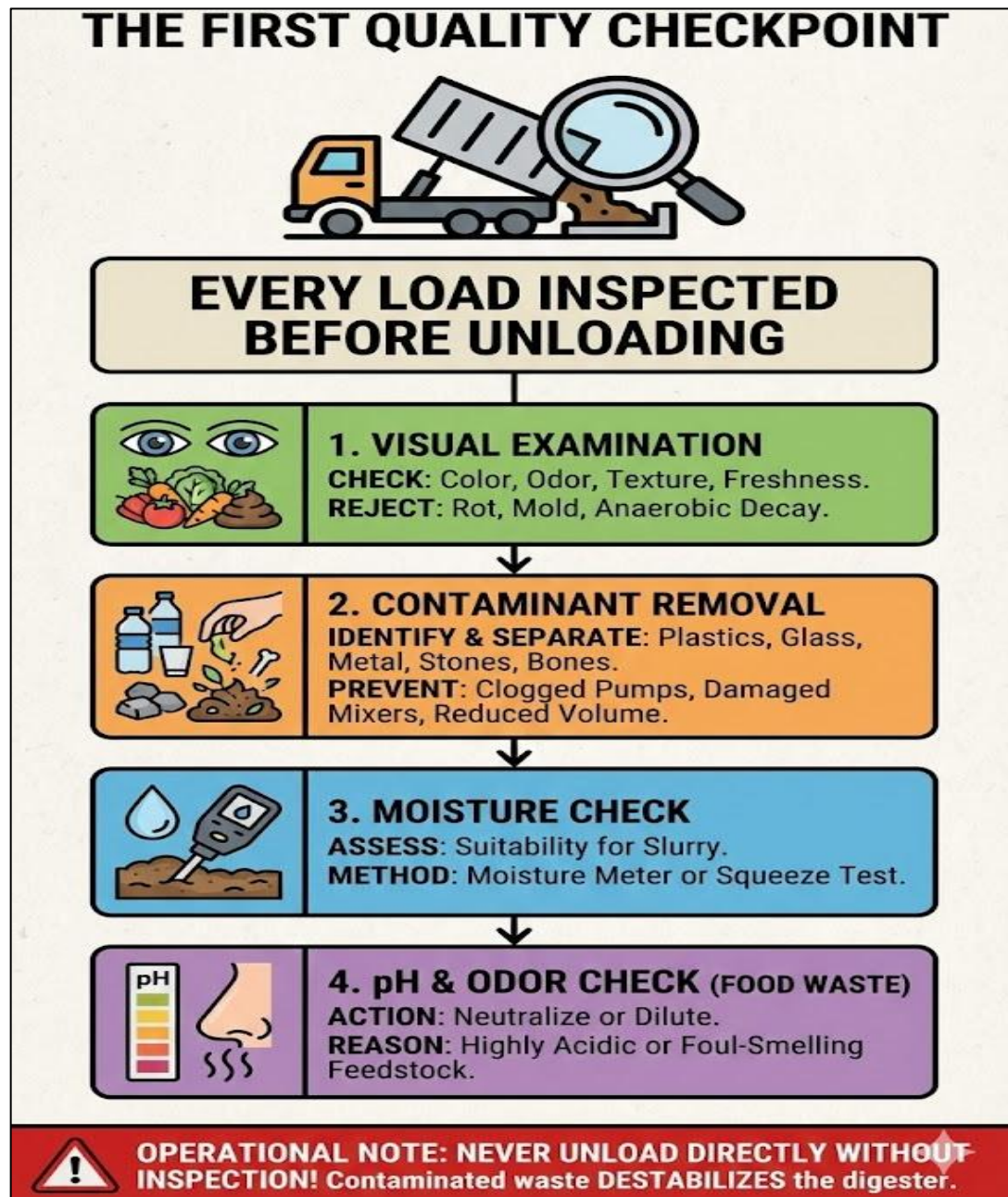


Figure 2: Initial Inspection and Quality Check of Feedstock

Unit 2.5 Feedstock Storage Management

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain why feedstock storage is a process-control activity, not a holding activity
- 2 Identify correct storage conditions for different feedstock types
- 3 Apply moisture, segregation, and FIFO principles during storage

Ask

- Ask the participants where feedstock is usually kept before feeding.
- Ask if they have experienced foul odour, heating, or gas drop after using stored feedstock.

Notes for Facilitation

- Many operators focus only on feeding and ignore storage behaviour.
- Emphasise that storage is where feedstock quality is either preserved or destroyed.
- Use examples of food waste fermenting overnight or straw heating internally.
- Reinforce that once volatile solids are lost in storage, they can never be recovered.

Table 2: Storage Conditions for Different Feedstock Types based on Moisture Content

Feedstock Type	Moisture (%)	Storage Condition
Crop residues	10–20	Aerated, raised from ground level
Animal manure	70–80	Shaded pit or covered tank
Food waste	70–90	Covered bins, use within 24–48 hrs
Press mud / pulp	50–75	Moisture-controlled silo

Explain

- Explain that after receiving and inspection, proper storage is the next critical step in feedstock management.
- Explain that storage determines whether energy value, moisture balance, and microbial suitability are preserved until feeding.

UNIT 2.6: Pre-treatment for Feedstock

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain why pre-treatment is essential before feeding material into the digester
- 2 Identify different types of feedstock pre-treatment and their purpose

Ask

- Ask the participants if they have ever seen long straw or fibrous material floating inside a digester.
- Ask whether gas production became uneven when unprocessed feed was added.
- Ask the participants if they know they can tell

Explain

- Explain that feedstock pre-treatment is the set of processes applied to raw organic material to make it suitable for anaerobic digestion.
- Explain that the goal is to improve biodegradability, enable uniform slurry formation, and maintain consistent digester performance.

UNIT 2.7: Operational Considerations and Safety while Pre-Treatment of Feedstock

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain why the pre-treatment zone is a high-risk, high-impact area in a biogas plant
- 2 Identify equipment-specific safety requirements during feedstock pre-treatment

Ask

- Ask the participants if they consider the pre-treatment area riskier than the digester area.
- Ask whether they have seen equipment damage or near-miss incidents during shredding or mixing.

Notes for Facilitation

- This section must be delivered slowly and seriously.
- Use real examples of jams, blade damage, or acid shocks where possible.

Explain

- Explain that the pre-treatment zone is one of the most active and hazardous operational areas in a biogas plant.
- Explain that here, organic materials interact with high-speed mechanical equipment and sometimes chemicals.
- Explain that even a small lapse, such as a stone entering a shredder or bypassing a safety

Unit 2.8 Health, Hygiene, and Safety Practices

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain why feedstock handling is a critical safety and process control stage
- 2 Identify safety risks associated with underfeeding and overfeeding
- 3 Apply safe loading, handling, and transfer practices during daily operations

Ask

- Ask whether they have seen gas fluctuations after irregular feeding.
- Ask the participants if they know they can tell
- that many safety incidents and digestion failures start with poor feeding discipline rather than equipment faults.

Notes for Facilitation

- Clarify that unsafe feeding practices stress both workers and microorganisms.
- Use examples from daily operations rather than rare accidents.
- Reinforce that most hazards are preventable through routine discipline.

Explain

- Explain that feedstock handling and loading lie at the intersection of plant safety and process reliability.
- Explain that every action at this stage affects microbial health, gas stability, and worker well-being.





3. Biogas Plant Components, Layout, and ancillary system

Unit 3.1: Fixed Dome Biogas Digester

Unit 3.2: Floating-Dome Biogas Digester

Unit 3.3: CSTR Biogas Digesters

Unit 3.4: Plug-Flow Biogas Digester

Unit 3.5: Operation and Maintenance of Biogas Ancillary Systems

Unit 3.6: Combined Heat and Power (CHP) Systems



Key Learning Outcomes



At the end of this module, you will be able to:

1. Explain the working principle of a fixed-dome biogas plant
2. Explain the working principle of a floating-dome biogas digester using hydraulic, biological, and pressure cycles.
3. Identify major components and systems such as the digester tank, mixing, heating, and instrumentation.
4. Identify key structural and functional components of a plug-flow digester and their roles.
5. Understand the role of ancillary systems in stable biogas and CBG plant operation.
6. Explain the role of CHP systems in biogas and CBG plants and their contribution to energy efficiency.
7. Identify major CHP components and understand their functions in power and heat generation.

Unit 3.1 Introduction to Industrial Fixed-Dome Biogas Plants

Unit Objectives

At the end of this unit, you will be able to:

- 1 Describe what an industrial fixed-dome biogas plant is and how it differs from other digester types.
- 2 Explain the basic structural and functional principles of a fixed-dome digester.

Ask

- Ask the participants if they have heard the term fixed-dome biogas plant before.
- Ask whether they know how gas is stored in a fixed-dome system without a separate gas holder.

Notes for Facilitation

- Start by giving an introduction about why fixed-dome designs are widely used in long-life biogas infrastructure.
- Tell participants that fixed-dome plants rely more on good civil construction quality than on mechanical equipment.

Explain

- An industrial fixed-dome biogas plant is based on a civil engineering design rather than mechanical complexity.
- The digester is constructed mainly from brick masonry and reinforced concrete, forming a sealed underground chamber.

Unit 3.1.1 Design and Structural Elements of Fixed-Dome Biogas Digesters

Unit Objectives

At the end of this unit, you will be able to:

- 1 Describe the structural design philosophy of fixed-dome biogas digesters.
- 2 Explain how shape, geometry, and underground construction support digestion and gas storage.

Ask

- Use these questions to link design with field experience.
- Ask the participants if they know why fixed-dome digesters are always circular and curved in shape.

Notes for Facilitation

- Start by giving an introduction about why fixed-dome digesters are civil structures, not mechanical systems.
- Tell participants that design mistakes cannot be corrected easily once construction is complete.

Explain

- Explain the design concept clearly and logically.
- A fixed-dome biogas digester is designed as a single underground civil structure that performs both digestion and gas storage.

UNIT 3.1.2: Hydraulic Components of Fixed-Dome Biogas Plants

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Describe the role of hydraulic components in a fixed-dome biogas plant.
- 2 Explain how slurry movement and gas pressure regulation occur without pumps or mechanical systems.

Ask

- Ask the participants if they know how slurry moves inside a fixed-dome plant without pumps.
- Ask whether they have seen slurry levels changing in inlet or outlet tanks during operation.

Encourage short responses and relate them to physical plant behavior.

Explain

- Hydraulic components govern slurry flow and gas pressure regulation in fixed-dome biogas plants.
- Unlike mechanically mixed systems, fixed-dome plants rely entirely on gravity and hydraulic displacement.

UNIT 3.1.3: Gas Storage and Pressure Management

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Describe how gas is stored in a fixed-dome biogas plant without mechanical holders.
- 2 Explain how gas pressure is regulated through hydraulic slurry displacement.

Ask

- Ask the participants if they know where biogas is stored in a fixed-dome plant.
- Ask whether they think gas pressure is controlled mechanically or naturally.

Explain

- In a fixed-dome biogas plant, gas storage and pressure control are achieved through structural design and hydraulic balance.
- The rigid dome, slurry movement, and gas pipeline together form a passive but reliable pressure management system.



Figure 3: Gas outlet pipe

UNIT 3.2: Introduction to Floating-Dome Biogas Digesters

Unit Objectives

At the end of this unit, participants will be able to:

- Describe the basic concept and working logic of floating-dome biogas digesters.
- Explain why the KVIC floating-dome model became widely adopted in India.

Ask

- Ask the participants if they have ever seen a biogas plant where the gas holder moves up and down.
- Ask whether they think gas pressure can be judged visually without gauges.

Explain

- The floating-dome biogas digester, commonly known as the KVIC model, is one of the most widely deployed biogas designs in India.
- It has been extensively used in households, farms, gaushalas, and small institutions where daily supervision is possible.

Notes for Facilitation

- Start by giving an introduction about why early biogas programmed in India preferred visible and easy-to-understand designs.
- Tell participants that the floating dome acts both as a gas holder and a pressure regulator.

UNIT 3.2.1: Structural Components of a Floating-Dome Biogas Plant

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Describe the key structural components of a floating-dome biogas plant.
- 2 Explain how civil construction replaces mechanical systems in maintaining stability.

Ask

- Ask the participants if a biogas plant can function reliably without mechanical pressure control.
- Ask whether poor construction can affect gas production even if feeding is correct.

Explain

- Explain how constant digester volume supports stable digestion.
- Tell them that misalignment of the dome can create friction and gas loss.



Figure 4: A Floating Dome Biogas Plant

UNIT 3.2.2: Anaerobic Digestion Process in Floating-Dome Systems

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the anaerobic digestion process occurring inside a floating-dome biogas digester.
2. Explain the four biochemical stages of anaerobic digestion and their sequence.

Ask

- Ask the participants if they know how long slurry stays inside a biogas digester.
- Ask whether biogas is produced instantly or through multiple biological steps.

Explain

- Explain that once slurry enters the digester, it undergoes a slow and carefully balanced biological transformation.
- Explain that floating-dome digesters are designed for a hydraulic retention time of about 40–60 days.

Biochemical Stages of Anaerobic Digestion

- **Hydrolysis**
 - Complex carbohydrates, proteins, and fats are broken into simpler soluble compounds.
- **Acidogenesis**
 - Soluble compounds are converted into volatile fatty acids, alcohols, and gases.
- **Acetogenesis**
 - Volatile fatty acids are converted into acetic acid, hydrogen, and carbon dioxide.
- **Methanogenesis**
 - Methanogenic bacteria convert acetic acid and hydrogen into methane-rich biogas.

Table 3: Typical Conditions Inside the Slurry Chamber

Parameter	Ideal Range	Role in Digestion
Temperature	25–35°C	Maximizes microbial activity
pH	6.8–7.2	Protects methanogens from acidity

UNIT 3.2.3: Operation and Maintenance Fundamentals

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the basic principles of operation and maintenance of a floating-dome biogas plant.
2. Explain how biological processes, hydraulics, and dome movement are interconnected.

Ask

- Ask the participants if they think a biogas plant needs instruments to indicate its condition.
- Ask whether dome movement and slurry behaviour can indicate internal problems.

Explain

- Explain that floating-dome biogas plants do not rely on automation or sensors.
- Explain that the plant communicates its condition through visible signs such as dome movement, slurry displacement, and flame quality.

Notes for Facilitation

- Start by giving an introduction about O&M as a discipline rather than a task list.
- Tell them that floating-dome plants “speak” through movement, smell, and flame.

UNIT 3.3: Introduction to CSTR Systems

Unit Objectives

At the end of this unit, you will be able to:

- 1 Explain the operating principles of CSTR systems.
- 2 Explain the CSTR systems

Explain

- Tell the learner about the working principle of Continuous Stirred Tank Reactor (CSTR) systems.
- Explain the continuous feeding and discharge process in the reactor.
- Discuss the role of agitators in maintaining uniform mixing inside the reactor.
- Explain the importance of temperature, pressure, and retention time in CSTR operation.

Notes for Facilitation

Tell the learner about the important operational aspects of CSTR systems:

- **Continuous Operation:** In a CSTR system, raw materials are continuously fed into the reactor while processed material is continuously removed.
- **Mixing Process:** Mechanical stirrers or agitators ensure uniform mixing of reactants, temperature, and concentration throughout the reactor.
- **Operating Conditions:** CSTR systems generally operate under controlled temperature and pressure conditions depending on the process requirement.
- **Retention Time:** The efficiency of the reactor depends on hydraulic retention time (HRT), which determines how long the material remains inside the reactor for reaction completion.

UNIT 3.3.1: Operating Parameters for CSTR Systems

Unit Objectives

At the end of this unit, you will be able to:

- 1 Describe the key operating parameters governing CSTR performance
- 2 Explain the interaction between HRT, OLR, temperature, and mixing

Ask

- Ask the participants if increasing gas production is possible by changing only one parameter
- Ask whether short HRT always improves plant output

Explain

- Explain that CSTR performance depends on parameter interaction, not single setpoints
- Explain the role of HRT in protecting slow-growing methanogens
- Explain why OLR is both a productivity driver and a stability risk

Notes for Facilitation

- Start by giving an introduction about operating envelopes in biological reactors
- Tell them that changing one parameter without adjusting others causes instability
- Tell them that HRT controls biological safety margins
- Tell them that OLR is the most common cause of digester failure when mismanaged
- Tell them that temperature stability alone is not enough without proper mixing
- Explain why gas pressure control is critical for downstream equipment
- List the various reasons operating discipline is essential in CSTR systems

UNIT 3.3.2: Operation and Maintenance of a CSTR Biogas Plant

Unit Objectives

At the end of this unit, you will be able to:

- Describe the importance of structured operation and maintenance in CSTR biogas plants
- Explain daily, weekly, monthly, and annual O&M activities for CSTR systems

Ask

- Ask the participants if minor neglect in daily operation can affect gas output
- Ask whether biological problems can start from mechanical issues such as poor mixing
- Ask if anyone has experienced gas drop due to skipped maintenance activities

Explain

- Explain that a CSTR is a controlled biological reactor, not a passive tank
- Explain that operation and maintenance follow a layered rhythm: daily, weekly, monthly, and annual

Notes for Facilitation

- Start by giving an introduction about why CSTR plants demand higher operational discipline than simple digesters
- Tell them that daily checks act as the first line of defense against failure
- Tell them that weekly maintenance prevents slow performance degradation
- Tell them that monthly inspections protect energy efficiency and gas quality
- Tell them that annual shutdowns restore long-term mechanical and control accuracy
- Explain that ignoring small deviations leads to compounded biological stress
- List the various reasons why structured O&M is essential in CSTR systems

UNIT 3.3.3: Troubleshooting of CSTR Biogas Plants

Unit Objectives

At the end of this unit, you will be able to:

- Describe early warning signals of instability in a CSTR biogas plant
- Explain how biological, process, and mechanical indicators are linked

Ask

- Ask the participants if a CSTR usually fails suddenly or gives advance warning
- Ask whether gas drop is always the first sign of trouble

Explain

- Explain that troubleshooting in a CSTR is a diagnostic process, not an emergency response
- Explain that CSTRs provide early and measurable warning signals

Notes for Facilitation

- Start by giving an introduction about why CSTR troubleshooting is proactive, not reactive
- Tell them that most failures are preceded by small, detectable deviations
- Tell them that operators must read trends, not isolated readings
- Tell them that biological stress often appears before visible gas reduction
- Tell them that mechanical overload is often a symptom, not the root cause
- Explain that calm, incremental correction is safer than aggressive adjustment
- List the various reasons why structured troubleshooting is critical in CSTR plants
 - Protects sensitive methanogens
 - Prevents acidification and souring
 - Avoids equipment damage due to overload
 - Maintains gas quality and pressure stability
 - Reduces unplanned shutdowns and energy loss

UNIT 3.4: Introduction to Plug-Flow Biogas Plants

Unit Objectives

At the end of this unit, participants will be able to

1. Describe the concept and working philosophy of plug-flow biogas plants
2. Explain how plug-flow digestion differs from fully mixed digester systems

Ask

- Ask the participants if they know the difference between a stirred tank digester and a plug-flow digester
- Ask whether they have seen or worked with long horizontal biogas reactors on dairy farms or gaushalas
- Ask what challenges they face while digesting fibrous or high-TS feedstock



Figure 5 : Exterior View of Plug-Flow Biogas Plant

Explain

- Explain the meaning of “plug flow” as unidirectional movement of slurry with minimal back-mixing
- Explain how digestion progresses naturally from inlet to outlet in stages

Notes for Facilitation

- Start by giving a simple real-world analogy such as material moving slowly through a pipeline or channel
- Tell them that plug-flow digesters rely more on reactor geometry than mechanical mixing
- Tell them about the typical horizontal or elongated structure of plug-flow reactors
- Tell them that these systems are widely used in dairy farms handling manure with bedding material
- Tell them about the operational advantage of lower energy consumption and fewer moving parts

Facilitator Tip

Emphasize that plug-flow technology is not a low-tech option but a purpose-driven design choice for specific feedstock conditions. Reinforce that selecting the right digester type is a design decision rooted in biology, not just equipment preference.

UNIT 3.4.1: Design and Structural Elements of Plug-Flow Biogas Plants

Unit Objectives

At the end of this unit, participants will be able to

1. Describe the core structural philosophy behind plug-flow biogas digesters
2. Explain how reactor geometry governs slurry movement and staged digestion

Ask (Engagement Questions)

- Ask participants what they think happens if slurry starts flowing faster in the centre than at the sides
- Ask if anyone has seen manure separating into layers and what problems that caused
- Ask why a long, narrow channel might behave more predictably than a round tank
- Ask what would happen if air entered the digester from the outlet side
- Ask whether adding mixers would improve or damage plug-flow behaviour
- Ask participants why dairy manure works better in plug-flow than food slurry



Figure 6 : Front cross-sectional view of Plug Flow biogas Plant

Explain (Core Concepts to Cover)

- Explain the concept of plug movement versus mixed movement
- Explain why minimal back-mixing is essential for staged digestion
- Explain how geometry replaces mechanical mixing in plug-flow systems
- Explain how each day's feeding acts like a piston pushing slurry forward

Notes for Facilitation (Detailed Guidance)

- Start by clearly stating that plug-flow digesters are civil-engineering-driven systems, not mechanical systems
- Tell participants that once a plug-flow reactor is built, its behaviour is largely fixed by geometry
- Tell them that poor design cannot be corrected later by operational adjustments
- Tell them that length-to-width ratio is more important than total volume
- Tell them about how even small shortcuts in wall finishing or slope create preferential flow paths
- Tell them that inlet jetting is one of the most common design failures in plug-flow plants
- Tell them that outlet chambers are not drains but hydraulic stabilisers
- Explain that the cover is not just a roof but part of the gas and thermal system
- Explain how gas pressure stability protects both digestion and downstream equipment
- Explain why insulation failures often show up first as methane loss, not temperature alarms
- Explain how fibrous feedstock benefits from staged digestion zones
- Explain why plug-flow systems tolerate solids but not hydraulic shocks
- List the various reasons plug-flow plants fail
 - Incorrect reactor proportions
 - Poor inlet transition design
 - Uneven floor slope or wall roughness
 - Inadequate insulation
 - Air leakage at outlet or cover joints

Trainer Emphasis Points

- Reinforce that structure is the process control system in plug-flow digesters
- Emphasise that plug-flow plants reward good construction more than advanced automation
- Clarify that simplicity does not mean forgiving design errors
- Highlight that plug-flow digesters are ideal where feedstock is thick, fibrous, and consistent
- Stress that civil precision is more important here than mechanical sophistication

Learning Reinforcement Prompt

This mental walkthrough helps participants internalise how structure defines biology in plug-flow systems.

UNIT 3.4.2: Construction Materials, Structural Integrity, and Waterproofing Systems

Unit Objectives

At the end of this unit, participants will be able to:

1. Describe the role of construction materials in ensuring long-term structural stability of plug-flow biogas digesters
2. Explain how internal surface finishes and protective coatings influence slurry movement and digestion behaviour

Ask

- Ask the participants if they have observed leakage, seepage, or wall damage in biogas digesters
- Ask whether they know how poor surface finishing can affect slurry movement in plug-flow systems
- Ask what could happen if groundwater enters a digester structure over time

Explain

- The importance of structural materials in withstanding slurry pressure, soil loads, and long-term thermal cycling
- How smooth internal finishes support true plug-flow behaviour and prevent fibre accumulation

Notes for Facilitation

- Start by giving an introduction about how plug-flow digesters rely on civil engineering rather than mechanical control
- Tell them that in plug-flow systems, the structure itself governs hydraulics, retention time, and stability
- Tell them about the long-term stresses acting on the reactor including slurry load, gas pressure, and temperature variation
- Tell them that construction defects cannot be corrected later through operational adjustments
- Tell them about the role of internal coatings in protecting concrete and maintaining smooth slurry movement

UNIT 3.4.3: Working Principles of a Plug-Flow Biogas Plant

Unit Objectives

At the end of this unit, participants will be able to:

1. Describe the fundamental working principle of plug-flow biogas digesters
2. Explain how slurry movement and reactor geometry replace mechanical mixing

Ask

- Ask the participants if they have seen or worked with long horizontal biogas digesters
- Ask whether they understand how slurry can move without mixers or recirculation

Explain

- That plug-flow digesters work without mechanical mixing or slurry recirculation
- That slurry moves forward as a cohesive plug due to high solids and viscosity
- How daily feeding gently displaces older material while preserving digestion order

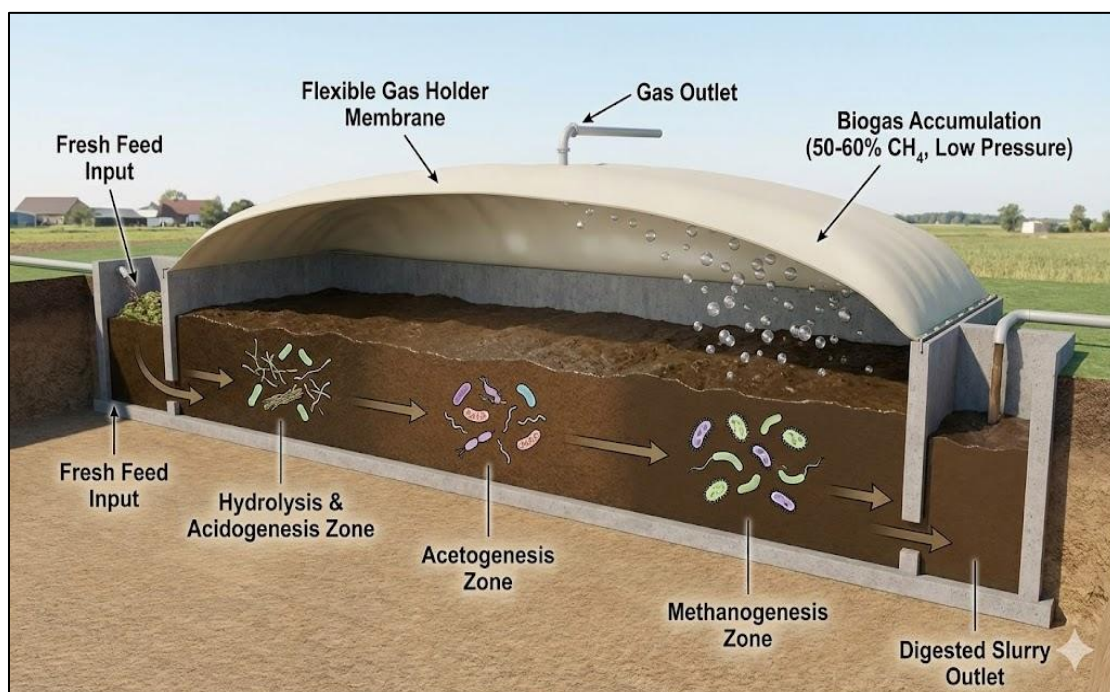


Figure 7: Working Cycles of Plug-Flow Digesters

UNIT 3.5: Introduction to Ancillary Systems in Biogas and CBG Plants

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the role of ancillary systems in biogas and CBG plant operation.
2. Explain how ancillary systems support gas handling, safety, and process stability.

Ask

- Ask the participants if they think plant reliability depends only on the digester and gas upgrading unit.
- Ask them what might happen if gas pipelines, storage, or pressure control systems fail even when digestion is healthy.

Explain

- Explain that these systems do not produce gas but determine whether gas can be safely handled, stored, purified, compressed, and delivered.
- Explain that pressure, flow, temperature, and gas quality are controlled primarily through ancillary systems.

Notes for Facilitation

- Start by giving an introduction about how ancillary systems quietly govern daily plant stability without being visibly prominent.
- Tell them that gas storage, pipelines, compressors, utilities, and safety devices operate in continuous interaction with the core process.
 - Maintain safe and stable gas flow and pressure
 - Protect upgrading, compression, and bottling equipment
 - Ensure regulatory compliance and operational safety
 - Prevent cascading failures across plant sections
 - Maximise plant uptime, efficiency, and asset life

UNIT 3.5.1: Gas Handling, Storage, and Leak Prevention Systems

Unit Objectives

At the end of this unit, you will be able to:

1. Describe what ancillary systems are and why they are essential in biogas and CBG plants.
2. Explain how ancillary systems support digestion, upgrading, compression, and dispatch operations.

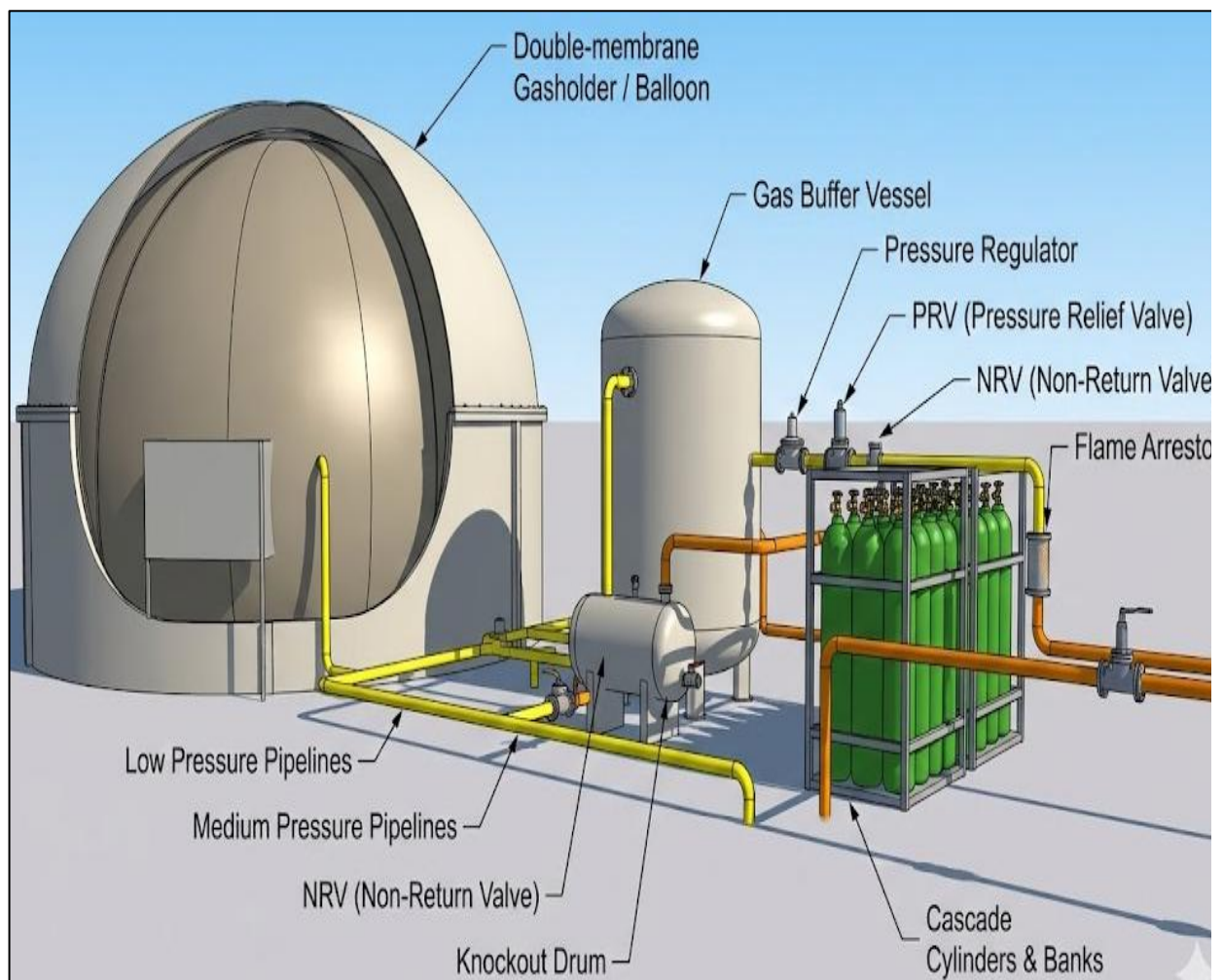


Figure 8: Gas Handling and Storage System in a Biogas Plant Setup

Ask

- Ask the participants if they know what systems operate *after* biogas leaves the digester.
- Ask them whether gas quality and pressure depend only on the digester or on supporting systems as well.

Explain

- Explain that ancillary systems are the supporting infrastructure that keeps the entire plant operable and safe.
- Explain that these systems do not produce gas but determine how effectively gas is handled, purified, compressed, and delivered.

Notes for Facilitation

- Start by giving an introduction about how a biogas plant is more than just a digester and a compressor.
- Tell them that ancillary systems form the connective tissue between biological, mechanical, and commercial operations.
 - They stabilise gas flow and pressure
 - They protect digesters and compressors from abnormal conditions
 - They enable reliable upgrading and compression
 - They maintain gas quality and safety
 - They determine overall plant reliability and uptime

UNIT 3.5.2: Biogas Purification and Upgrading Systems

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the purpose of biogas purification and upgrading systems in a biogas and CBG plant.
2. Explain how impurities such as CO₂, H₂S, moisture, and particulates affect gas quality and equipment reliability.

Ask

- Ask the participants if they know why raw biogas cannot be directly compressed or injected without purification and upgrading.

Explain

- Explain that purification and upgrading systems act as the quality gate between digestion and compression, determining whether biogas becomes a usable, safe, and market-ready fuel.

Team Activity

- Identify the role of purification and upgrading in biogas quality improvement.
- Discuss common impurities removed during purification.
- Explain how purification affects safety and gas usability.

Notes for Facilitation

- Start by giving an introduction about the difference between raw biogas and fuel-grade methane.
- Tell them that raw biogas contains carbon dioxide, hydrogen sulphide, moisture, and fine particles that lower calorific value and damage downstream equipment.

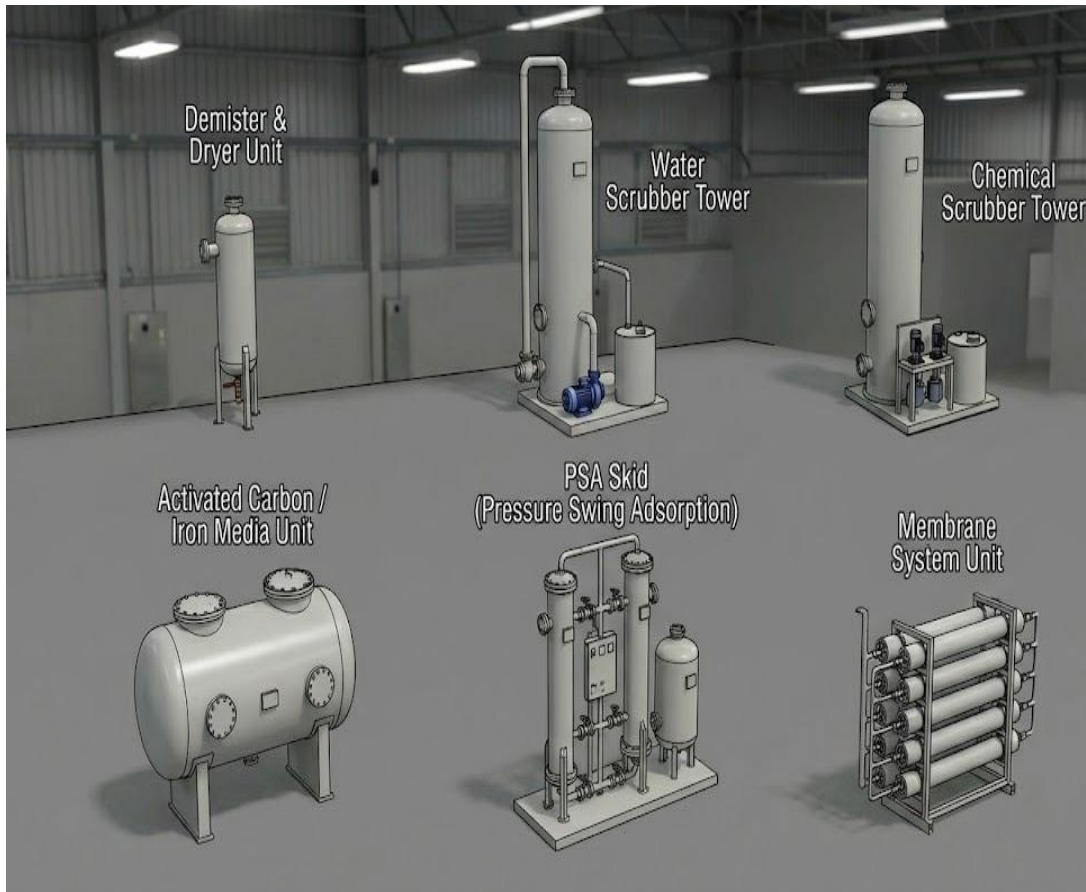


Figure 9: Biogas Purification and Upgrading Systems in a Biogas Plant Setup

UNIT 3.6: Overview of CHP Systems in Biogas and CBG Plants

Unit Objectives

1. Describe what a Combined Heat and Power (CHP) system is and where it is used in biogas and CBG plants.
2. Explain how CHP systems convert biogas into electricity and usable heat simultaneously.

Ask (Participant Engagement)

- Ask the participants if they know the meaning of *Combined Heat and Power* or *cogeneration*.
- Ask whether they have seen or worked near a CHP engine in a biogas or CBG plant.
- Ask what happens to heat in a normal power generator and whether that heat is usually used or wasted.

Explain (Concept Delivery)

- Explain that a CHP system is an engine or generator that produces electricity while capturing the heat generated during combustion.
- Explain that in biogas plants, this recovered heat is not waste but a valuable input used to maintain digester temperature.

Notes for Facilitation

- Start by giving a simple, non-technical introduction to energy use in a biogas plant.
- Tell participants that CHP is different from a normal diesel generator because it uses both power and heat.

UNIT 3.6.1: Major Energy Uses in a Biogas Plant Powered by CHP

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the main electrical and thermal energy demands in a biogas and CBG plant.
2. Explain how CHP-generated electricity supports continuous plant operation

Ask (Participant Engagement)

- Ask the participants if they know which equipment in a biogas plant consumes the most electricity.
- Ask whether heat produced by engines or generators is usually wasted or reused in plants they have seen.

Explain (Concept Delivery)

- Explain that the CHP system acts as the central energy source for both power and heat in the plant.
- Explain that electricity from CHP runs mixers, agitators, pumps, blowers, separators, lighting, PLC panels, and SCADA systems.

Notes for Facilitation

- Start by giving an introduction about how energy flows through a biogas plant.
- Tell them that almost every critical operation depends on reliable electricity supply.

UNIT 3.6.2: Working Principle of CHP Systems

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the working principle of a Combined Heat and Power (CHP) system.
2. Explain how biogas is converted into electrical and thermal energy.

Ask (Participant Engagement)

- Ask the participants if they know what happens to heat in a normal power generator.
- Ask whether electricity generation alone represents full energy recovery from fuel.

Explain (Concept Delivery)

- Explain that a CHP system converts biogas into electricity and heat at the same time.
- Explain that biogas is combusted in a gas engine, producing mechanical energy.

Notes for Facilitation

- Start by giving an introduction about energy conversion in conventional power plants.
- Tell them that CHP differs because it does not waste heat.
- Tell them about the two energy outputs of CHP: electrical energy and thermal energy.

UNIT 3.6.3: Common CHP Operational Problems

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the typical operational problems observed in CHP systems.
2. Explain how gas quality, combustion, mechanical condition, and electrical loading are interlinked.

Ask (Participant Engagement)

- Ask the participants if CHP failures usually happen suddenly or develop over time.
- Ask whether engine trips always indicate a single root cause.

Explain (Concept Delivery)

- Explain that most CHP operational problems evolve gradually rather than occurring instantly.
- Explain that issues often start in one area, such as gas quality, and then cascade into combustion, mechanical, and electrical problems.

Notes for Facilitation

- Start by giving an introduction about CHP as an integrated system rather than separate components.
- Tell them that ignoring small deviations often leads to large failures.
- Tell them that gas quality issues frequently trigger downstream engine and electrical problems.



4. Monitoring of Biogas Plants

UNIT 4.1: Introduction to Monitoring

UNIT 4.2: Standard Operating Procedures (SOPs) for Biogas Production

UNIT 4.3: Sampling Protocols

Unit - 4.4 Digestate Management & Utilization

UNIT 4.5: Advanced Monitoring and Laboratory Parameters

UNIT 4.6: Compliance, Environmental Monitoring, Preventive & Corrective Maintenance

UNIT 4.7: Troubleshooting, Safety, Compliance & Emergency Protocols

UNIT 4.8: Seasonal O&M Adjustments

UNIT 4.9: Troubleshooting SOPs, Safety, PPE & Emergency Protocols



Key Learning Outcomes

At the end of this module, you will be able to:

- 1 Explain key monitoring parameters that indicate digester health and plant stability.
- 2 Apply SOPs for feeding, gas handling, sampling, and digestate management in daily operations.
- 3 Carry out correct sampling and testing of feedstock, slurry, digestate, and biogas.
- 4 Maintain safety, compliance, and readiness for abnormal and emergency operating conditions.

UNIT 4.1: Introduction to Monitoring

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Describe why monitoring is a critical operational function in biogas and CBG plant O&M
 2. Explain how monitoring parameters reflect digester health, process stability, and equipment performance

Ask – Participant Engagement (Opening Interaction)

- Begin the session by engaging participants with simple, experience-based questions.
- Ask the participants:
- Do you know what “monitoring” means in a biogas plant

Notes for Facilitation

- Start by giving an introduction about monitoring
- Explain that monitoring is the operator’s way of observing an invisible process.

Tell them

- Tell participants that monitoring helps operators understand whether the plant is running safely, efficiently, or moving toward trouble.
- Small changes in parameters usually appear days before major failures.
- Emphasise that most serious problems do not happen suddenly. They develop slowly and show early warning signs through monitored values.

Explain monitoring from an O&M perspective

- Explain that from an O&M point of view, monitoring enables:
 - -Early detection of acidification and foaming
 - -Identification of gas yield reduction trends
 - -Prevention of compressor, blower, or engine trips
 - -Detection of unsafe gas conditions such as high H₂S or air ingress
- Monitoring allows operators to act gradually and safely rather than reacting during emergencies.

UNIT 4.2: Standard Operating Procedures (SOPs) for Biogas Production

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe the role of Standard Operating Procedures in ensuring stable digestion, safe operations, and predictable performance in biogas and CBG plants.
 2. Explain the standard operating procedures for feeding, gas collection and safety, and digestate or slurry removal, and their importance in day-to-day operation and maintenance activities.

Ask

- Ask the participants if they know the meaning of Standard Operating Procedure (SOP).
- Then ask follow-up questions such as:
- Why do similar biogas plants sometimes perform very differently

Explain

- Explain that a Standard Operating Procedure is a written and agreed method that defines how a task must be performed every time, irrespective of who the operator is or which shift is running. In biogas plants, SOPs are especially important because the core process is biological, slow-responding, and invisible to the human eye.
- Explain that SOPs convert experience, best practices, and safety requirements into repeatable actions, ensuring that plant performance does not depend on individual judgement alone.

Notes for Facilitation

- Start by giving an introduction about SOPs in biogas plants
- Start by explaining that Standard Operating Procedures are the foundation of predictable and safe biogas plant performance. They define how operators must execute critical tasks in a consistent and controlled manner so that the biological system inside the digester remains stable over long periods.
- Explain that anaerobic digestion takes place inside a sealed reactor, where operators cannot directly see microbial activity. Because of this, SOPs become the only reliable way to ensure that external actions do not disturb internal biological balance.



Figure 10: Protocol to be followed for Flare Operation

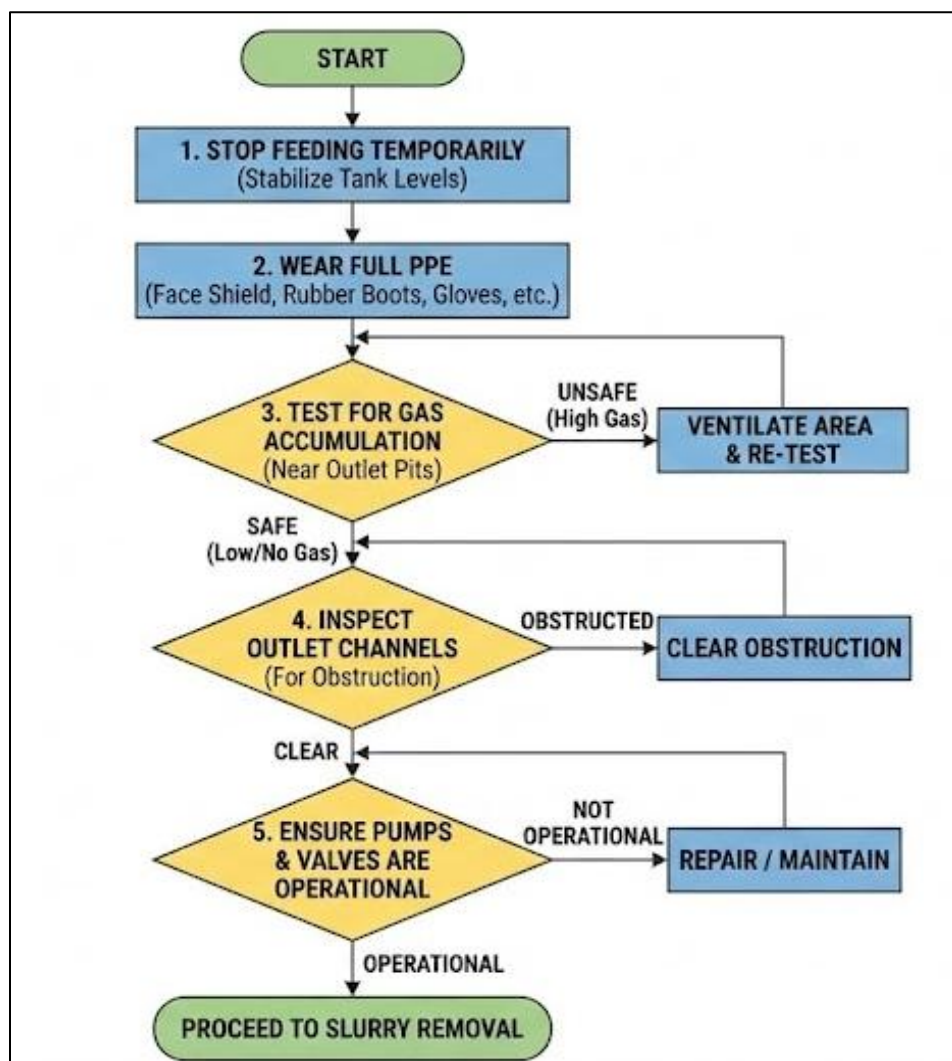


Figure 11: Steps to be followed during digestate pre-removal

UNIT 4.3: Sampling Protocols

Unit Objectives

- At the end of this unit, you will be able to:
- 1. Describe the purpose and importance of systematic sampling in biogas and CBG plants for monitoring digester health, process stability, and gas quality.
- 2. Explain the correct sampling protocols for feedstock, slurry, digestate, and biogas, including storage, labelling, moisture testing, and laboratory analysis.

Ask

- Then ask probing questions such as:
- -Can we see what is happening inside a digester
- -How do we know if microbes are healthy or stressed

Explain

- Explain that sampling is the only window into the digester. Since anaerobic digestion happens inside a sealed and opaque reactor, operators cannot visually observe microbial activity, acid build-up, or conversion efficiency.
- Explain that correct sampling allows operators to:
- -understand biological health

Notes for Facilitation

- Start by giving an introduction about sampling in biogas plants
- Start by explaining that sampling is one of the most critical diagnostic and quality control activities in biogas and CBG plants. Every operational decision related to loading, dilution, gas safety, and troubleshooting depends on the quality of samples collected.
- Explain that sampling is not a laboratory activity alone; it is an operational discipline that begins at the plant floor.

Unit - 4.4 Digestate Management & Utilization

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Describe digestate as a key output of biogas and CBG plants alongside biogas
 2. Explain the role of digestate in nutrient recycling, soil health improvement, and plant economics
 3. Identify risks associated with improper digestate handling and storage

Ask

- Ask the participants if they see digestate as a waste product or a valuable resource.
- Ask them whether they have observed issues such as odour complaints, slurry overflow, or storage problems related to digestate handling.

Explain

- Explain that digestate is the second major output of a biogas or CBG plant and, when managed correctly, contributes significantly to economic viability and sustainability.
- Explain that digestate contains valuable nutrients and stabilised organic matter that support soil fertility, reduce chemical fertiliser demand, and enable sustainable agriculture.
- Explain that digestate is often misunderstood and mishandled, leading to odour problems, nutrient losses, groundwater contamination, and environmental non-compliance.
- Explain that this figure shows the controlled discharge of digestate slurry from the biogas system into a storage or post treatment tank.

Notes for Facilitation

- Start by clarifying that digestate is a product, not a waste
- Tell them that most community complaints arise from poor digestate handling rather than gas systems
- Tell them that digestate management links plant operation with agriculture and environmental compliance

UNIT 4.4.1 - Nature and Composition of Digestate

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Describe digestate as a biologically stabilised output of anaerobic digestion
 2. Identify the three main fractions of digestate and their characteristics
 3. Explain how digestate composition influences handling, storage, and utilisation

Ask

- Ask the participants if all digestate looks and behaves the same.
- Ask them whether liquid digestate and solid digestate can be handled or applied in the same way.

Explain

- Explain that digestate is a biologically stabilised material produced after anaerobic digestion and is inherently non homogeneous.
- Explain that digestate separates naturally or mechanically into three fractions: liquid, solid, and stabilised or cured digestate.

Notes for Facilitation

- • Start by explaining why digestate is non homogeneous
- • Tell them that misuse often happens when fractions are not distinguished
- • Tell them that liquid digestate behaves like a nutrient solution while solids behave like organic matter
- • Explain that stabilised digestate has the highest acceptance for direct field application.

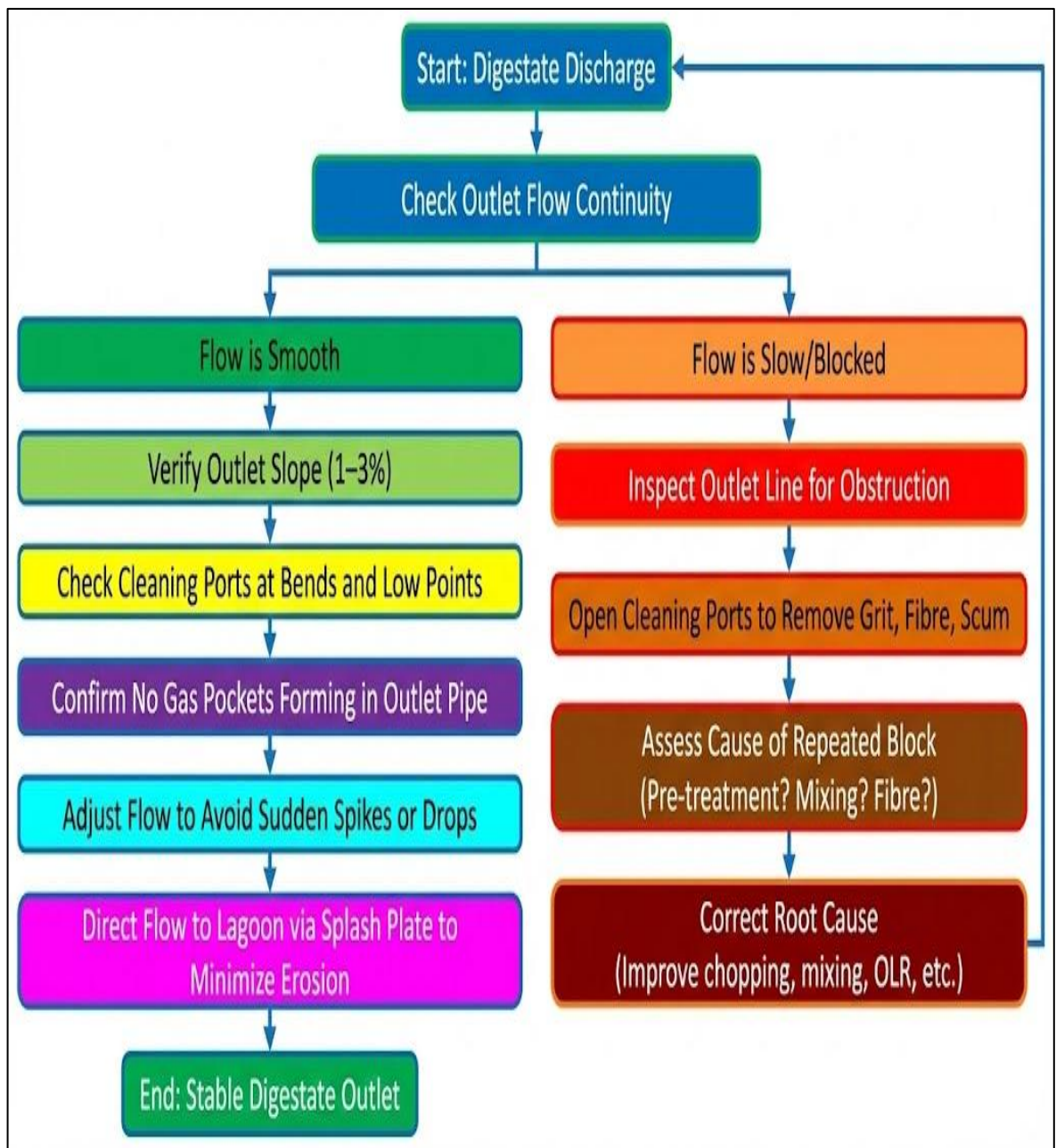


Figure 12: SOP on Digestate Outlet Management

UNIT 4.4.2: Digestate Properties Affecting O&M Decisions

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Identify key physical and chemical properties of digestate relevant to plant operations
 2. Explain how digestate properties influence storage, dewatering, stabilisation, and utilisation decisions
 3. Use digestates characteristics as operational indicators for corrective action

Ask

- Ask the participants if all digestate can be handled, stored, and applied in the same way.
- Ask them whether they have experienced odour, storage overflow, or low market acceptance of digestate and what they think caused these issues.

Explain

- Explain that digestate quality directly affects fertiliser value, environmental performance, and overall operational stability of a biogas or CBG plant.
- Explain that digestate is produced continuously and stored in large volumes, so even small quality issues can scale into major operational challenges.

UNIT 4.5: Advanced Monitoring and Laboratory Parameters

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Explain why laboratory monitoring is essential in anaerobic digestion
 2. Interpret key parameters as diagnostic indicators
 3. Prevent operational failures through early intervention

Explain

- Explain that digestion occurs inside a sealed, oxygen-free system and cannot be observed directly
- Explain that laboratory parameters act like health indicators, revealing microbial balance, substrate quality, and process stress.
- Explain that data-driven monitoring transforms plant operation from guesswork to predictive control.

Notes for Facilitation

- Emphasise trend analysis over single values
- Tell them monitoring enables prevention, not firefighting
- Link lab data to daily O&M decisions
- This section completes the digestate and monitoring module by linking field utilisation, compliance discipline, and laboratory diagnostics into a single, integrated O&M control framework for biogas and CBG plants.

UNIT 4.6: Compliance, Environmental Monitoring, Preventive & Corrective Maintenance

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe the importance of regulatory compliance and environmental monitoring in ensuring safe, lawful, and socially acceptable biogas and CBG plant operation
 2. Explain how preventive and corrective maintenance protect digester biology, mechanical systems, environmental performance, and long-term plant reliability

Ask

- Ask the participants whether a biogas or CBG plant producing stable gas output is automatically compliant and safe.
- Ask them what happens if monitoring data, maintenance logs, or safety checks are missing during an inspection or incident investigation.

Explain

- Explain that operating a biogas or CBG plant is not limited to producing gas. It requires continuous control over environmental emissions, equipment condition, worker safety, and regulatory obligations.
- Explain that every operational stage, from feedstock reception to digestion, gas purification, compression, and digestate handling, directly interacts with air, water, soil, and surrounding communities.

Notes for Facilitation

- Start by emphasising that compliance is about permission to operate, not paperwork
- Tell them monitoring prevents problems before regulators intervene
- Tell them maintenance protects microbes as much as machines

UNIT 4.6.1: Regulatory Compliance Framework

Unit Objectives

- At the end of this unit, you will be able to:
 1. Identify the regulatory authorities governing biogas and CBG plant operation
 2. Explain key compliance areas and the importance of documentation

Ask

- Ask participants whether MNRE approval alone allows unrestricted plant operation.
- Ask them which authority can immediately stop plant operations in case of gas safety or environmental violations.

Explain

- Explain that biogas and CBG plants operate under a multi-regulatory framework.
- Explain that each authority focuses on a specific risk area such as environmental protection, gas safety, electrical systems, or worker welfare.
- Explain that compliance is cumulative. Failure to meet even one authority's requirement can override approvals from others.

Notes for Facilitation

- Emphasise that undocumented compliance is legally treated as non-compliance
- Tell them inspectors trust records more than explanations

UNIT 4.6.2: Environmental Monitoring Requirements

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe the role of environmental monitoring in compliant plant operation
 2. Identify core environmental parameters and monitoring methods

Ask

- Ask participants how they would detect environmental harm without monitoring data.
- Ask them whether odour complaints usually appear before or after regulatory action.

Explain

- Explain that environmental monitoring acts as an early warning system.
- Explain that it allows operators to detect deviations before they escalate into violations or community complaints.

UNIT 4.6.3: Monitoring of Methane Leakage

Unit Objectives

- At the end of this unit, you will be able to:
 - Explain why methane leakage monitoring is critical
 - Identify common leak points and apply corrective actions

Ask

- Ask participants whether methane leaks always produce smell or visible signs.
- Ask them how small leaks affect efficiency and safety over time.

Explain

- Explain that methane is both the primary fuel and a powerful greenhouse gas.
- Explain that even minor leaks reduce energy efficiency, distort performance data, and increase explosion risk.

Notes for Facilitation

- Emphasise methane loss as both economic and safety risk
- Tell them leaks grow if ignored

UNIT 4.7: Troubleshooting, Safety, Compliance & Emergency Protocols

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe a structured troubleshooting approach for biogas and CBG plant operations
 2. Explain the link between safety, compliance, and effective fault resolution

Ask

- Ask the participants whether troubleshooting starts after a breakdown or before it.
- Ask them if reacting quickly without understanding the cause always solves the problem.

Explain

- Explain that operating a biogas or CBG plant requires more than technical knowledge. It requires disciplined thinking, safe behaviour, and compliance awareness.
- Explain that troubleshooting and safety are inseparable. Every fault occurs in an environment involving methane, hydrogen sulphide, pressurised systems, rotating equipment, electrical panels, and confined spaces.
- Explain that unsafe troubleshooting creates secondary incidents such as gas leaks, fires, toxic exposure, or digester damage.

Notes for Facilitation

- Emphasise that safety isolation comes before fault correction
- Tell them that biological systems respond slowly and must never be rushed
- Explain that documentation closes the troubleshooting loop
- Reinforce that repeated faults usually mean the root cause was not addressed

UNIT 4.8: Seasonal O&M Adjustments

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe how seasonal changes affect biogas and CBG plant performance
 2. Explain why seasonal adjustments are a preventive O&M requirement rather than a corrective response

Ask

- Ask the participants whether a plant that runs well in one season will automatically perform well in all seasons.
- Ask them which parameter they think is most affected by seasonal change—temperature, feedstock quality, moisture, or microbial activity.

Explain

- Explain that biogas production is a biological process and is therefore highly sensitive to environmental conditions.
- Explain that temperature, moisture, feedstock composition, and handling practices change across winter, summer, and monsoon seasons, directly affecting digestion stability and gas yield.
- Explain that seasonal O&M adjustments are part of preventive maintenance. Plants that wait for gas drop or instability before acting usually lose weeks of performance.

Notes for Facilitation

- Emphasise that seasonal problems are predictable
- Tell them recovery from seasonal damage is slower than prevention
- Explain that seasonal discipline protects microbes, not just machines
- Reinforce that seasonal O&M planning reduces downtime and complaints

UNIT 4.9: Troubleshooting SOPs, Safety, PPE & Emergency Protocols

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe the standard operating procedure for troubleshooting in biogas and CBG plants
 2. Explain why safety protocols and PPE are mandatory before any troubleshooting activity

Ask

- Ask the participants whether troubleshooting should begin with parameter adjustment or diagnosis.
- Ask them what risks arise when operators approach gas lines or digesters without safety preparation.

Explain

- Explain that troubleshooting in a biogas or CBG plant is a structured assessment of deviations identified through SCADA alarms, sensor readings, laboratory trends, or operator observations.
- Explain that most operational problems present predictable symptoms such as pressure variation, pH drift, unusual noise, odour release, or equipment trips.

Notes for Facilitation

- Emphasise diagnosis before adjustment
- Tell them that most repeat failures come from incomplete troubleshooting
- Reinforce documentation as part of troubleshooting, not an afterthought





Key Learning Outcomes

At the end of this module, you will be able to:

- 1 Identify major hazards in biogas plants, including H₂S, methane leaks, fire risks, and slurry contamination.
- 2 Use and maintain essential PPE such as masks, gloves, boots, and SCBA sets.
- 3 Follow hygiene practices required for waste-handling and biogas plant operations.
- 4 Perform basic gas leak detection and respond to early warning signs.
- 5 Apply first-aid and emergency procedures for fire, gas exposure, and chemical contact.
- 6 Execute emergency shutdown steps for leaks, overheating, or high digester pressure.

UNIT 5.1: Introduction to Health and Hygiene Fundamentals in Biogas Environments

Unit Objectives

1. Describe a biogas plant as an integrated end to end biological and mechanical process system
2. Explain why health and hygiene are critical for digester stability, operational reliability, and worker safety
3. Identify hygiene related risks associated with feedstock handling, digestion, gas processing, and digestate management

Ask

- Ask the participants if they know the meaning of health and hygiene in an industrial biogas or CBG plant context.
- Encourage them to share examples from their own experience such as spills, odour, blocked drains, dirty work areas, or health issues faced during plant operation.

Explain

- Explain that a biogas plant functions as a tightly integrated process system where each stage directly influences the next.

Notes for Facilitation

- Start by giving an introduction to the biogas plant as a complete process system rather than isolated equipment

Unit 5.1.1: Biogas Plant as an Integrated Process System

Unit Objectives

At the end of this unit, participants will be able to:

- Describe a biogas plant as an integrated biological and mechanical process system
- Explain how disturbances in one section of the plant affect downstream operations

Ask

- Ask the participants if they think problems in one area of the plant can affect other sections.

Explain

- Explain that a biogas plant does not operate as isolated equipment but as a continuous end to end process.

Notes for Facilitation

- Start by explaining the plant using a simple process flow from feedstock entry to digestate exit

UNIT 5.1.2: Hygiene as an Operational Safeguard

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain hygiene as a preventive operational safeguard in biogas and CBG plants
2. Identify routine hygiene practices that reduce safety, biological, and environmental risks

Ask

- Ask the participants if hygiene in a biogas plant only means cleaning floors.

Explain

- Explain that hygiene in a biogas or CBG plant is embedded into daily operating routines, not treated as an occasional cleaning task.

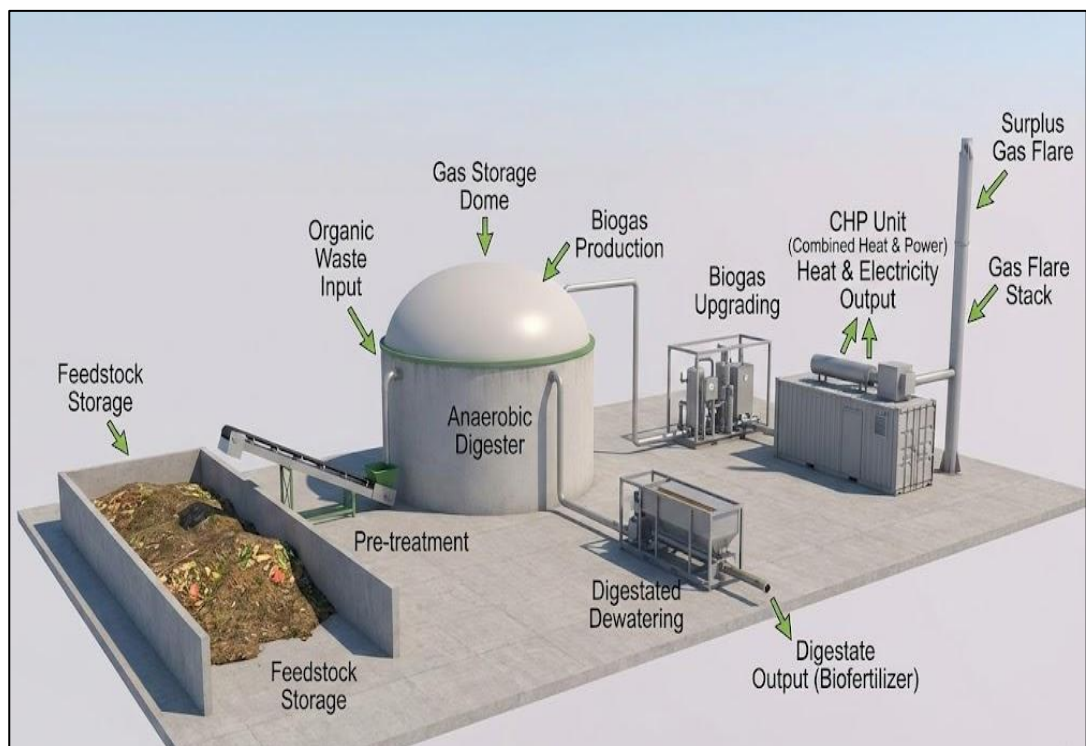


Figure 13: Biogas Production, Upgradation and CHP Step

UNIT 5.2: Importance of Hygiene in Biogas Operations

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain hygiene as a core operational requirement in biogas and CBG plants
2. Understand how hygiene influences digester stability, worker safety, and gas yield

Ask

- Ask the participants why hygiene is more critical in biogas plants than in conventional power systems.

Explain

- Explain that hygiene plays a central role in safe, stable, and efficient biogas and CBG plant operation.

Notes for Facilitation

- Emphasise that hygiene failures often appear first as process or safety issues

UNIT 5.2.1: Worker Health Protection

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Identify major occupational health risks in biogas and CBG plant operations
- 2 Explain how hygiene practices reduce exposure pathways

Ask

- Ask the participants what types of health problems are most common in waste and slurry handling environments.

Explain

- Explain that operators in biogas and CBG plants work in close contact with slurry, digestate, odorous gases, bioaerosols, and chemical residues on a daily basis.

Notes for Facilitation

- Emphasise that health protection starts with daily habits

UNIT 5.2.2: Prevention of Cross-Contamination Across Plant Zones

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Explain how cross-contamination occurs between different plant zones
- 2 Identify common contamination transfer pathways in biogas and CBG plants

Ask

- Ask the participants if contamination only spreads through pipelines and process flows.

Explain

- Explain that biogas and CBG plants are organised into distinct operational zones, each with a different hygiene risk profile.

Notes for Facilitation

- Emphasise that people are the main carriers of contamination

UNIT 5.2.3: Maintaining Environmental Cleanliness

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Explain environmental cleanliness as a core indicator of good O and M practice
- 2 Identify high risk plant zones where cleanliness directly affects safety and operations

Explain

- Explain that environmental cleanliness is both visible and measurable and reflects the overall discipline of plant operation.
- Explain that a clean plant is easier to operate, safer to audit, and more reliable over long-term operation.

UNIT 5.2.4: Hygiene Risks Across Plant Areas

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Identify hygiene risks present across different zones of a biogas or CBG plant
- 2 Understand how operator behaviour influences the spread or control of contamination

Ask

- Ask the participants if hygiene risks are limited only to feedstock or digestate areas.

Explain

- Explain that hygiene risks exist across all zones of a biogas or CBG plant, not only where waste or digestate is handled.
- Explain that when these practices are ignored, contaminants move from dirty zones to clean zones, increasing safety risks, odour issues, corrosion, and biological disturbances.

Notes for Facilitation

- Emphasise that hygiene failures usually spread through people, not pipes

UNIT 5.3: Operator Hygiene and Personal Cleanliness

Unit Objectives

At the end of this unit, participants will be able to:

- 1 Explain why operator hygiene is a critical operational control in biogas and CBG plants
- 2 Identify how personal cleanliness affects worker health, plant safety, and digester stability

Ask

- Ask the participants if operator hygiene only protects the individual or also protects the plant.

Explain

- Explain that operator hygiene and personal cleanliness are foundational to safe and reliable biogas and CBG plant operation.
- Explain that operators routinely handle organic feedstock, digestate, chemicals, and process equipment that are biologically active and potentially hazardous.

Notes for Facilitation

- Emphasise that hygiene protects both people and process

UNIT 5.3.1: Personal Hygiene Responsibilities of Operators

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain personal hygiene as a continuous operational responsibility
2. Identify daily hygiene practices required across all plant zones

Ask

- Ask the participants whether personal hygiene is only about self protection or also about protecting the process and coworkers.

Explain

- Explain that personal hygiene is a continuous responsibility that applies throughout the entire work shift and across all operational zones of a biogas or CBG plant.

Notes for Facilitation

- Emphasise that hygiene failures usually spread through people, not equipment
- Tell them that discipline in small habits prevents major safety issues
- Explain that clean operators protect clean processes
- Reinforce that consistent hygiene reflects professionalism and operational maturity

PERSONAL CLEANLINESS REQUIREMENTS FOR WASTE HANDLING



Figure 14: Personal Cleanliness Requirements

UNIT 5.3.2: Handwashing Discipline and Sanitation Facilities

Unit Objectives

At the end of this unit, participants will be able to:

- Explain why handwashing is a critical hygiene control in biogas and CBG plants

Ask

- Ask the participants how infections or contamination usually enter the human body in industrial environments.

Explain

- Explain that handwashing discipline is one of the most effective and non-negotiable hygiene controls in biogas and CBG plant operations.
- Explain that hand hygiene also protects clean zones such as control rooms, offices, and electrical panels from biological and chemical contamination.
- Continuous running water

UNIT 5.3.3: Hygiene Standards in Work Areas

Unit Objectives

At the end of this unit, participants will be able to:

- Identify how cleanliness supports process stability and equipment protection
- Apply structured cleaning routines to maintain professional O&M standards

Ask

- Ask the participants how quickly they can identify a leak or abnormal condition in a cluttered or dirty plant.

Explain

- Explain that hygiene standards in work areas form the foundation of safe, reliable, and well controlled biogas and CBG plant operations.
- Explain that regular cleaning ensures these deviations are noticed early and corrected before they lead to equipment damage, downtime, or safety incidents.
- Lower fire and electrical risk

Notes for Facilitation

- Emphasise that cleanliness improves fault detection speed
- Tell them that most accidents happen in neglected areas
- Explain that hygiene standards reflect operational maturity
- Reinforce that every operator is responsible for the cleanliness of their work area

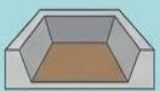
AREA	CLEANING REQUIREMENT	FREQUENCY
 Walkways	 Remove slurry, mud, spills	 Multiple times daily
 Feedstock Pits	 Remove residues, wash sides	 Daily
 Digestate Areas	 Keep drains open, wash edges	 Daily
 Control Rooms	 Keep PPE outside, dust-free area	 Daily
 Tools & Equipment	 Wash after use, disinfect where required	 Shift end

Figure 15: Facility Daily Cleaning Schedule

UNIT - 5.3.4: Odour, Pest, and Vector Control Practices

Unit Objectives

At the end of this unit, participants will be able to:

- Explain odour as an early operational warning signal rather than a cosmetic issue
- Identify conditions that promote pest and vector infestation in biogas and CBG plants

Ask

- Ask the participants whether odour problems usually appear suddenly or develop gradually.
- Explain
- Explain that in biogas and CBG plant operations, odours are not merely a nuisance but an early indicator of uncontrolled microbial activity, material accumulation, or inadequate sanitation.

Notes for Facilitation

- Emphasise that odour is an early warning, not the final problem

UNIT 5.4: Hygiene Infrastructure and Regulatory Compliance

Unit Objectives

At the end of this unit, you will be able to:

1. Describe why hygiene infrastructure is a regulatory and operational requirement in biogas and CBG plants
2. Explain how hygiene infrastructure supports compliance with environmental, health, and safety norms

Ask

- Ask the participants whether hygiene is checked only during inspections or continuously during operation.

Explain

- Explain that adequate hygiene infrastructure is not optional in biogas and CBG plants.

Notes for Facilitation

- Emphasise that inspectors trust what they see before what they read

UNIT 5.4.1: Minimum Hygiene Infrastructure Requirements

Unit Objectives

At the end of this unit, you will be able to:

- Describe the minimum hygiene infrastructure required in biogas and CBG plants
- Explain how each hygiene infrastructure element supports safe and compliant operations

Ask

- Ask the participants if hygiene rules can be followed without proper facilities.

Explain

- Explain that minimum hygiene infrastructure forms the baseline for safe, compliant, and efficient operation of biogas and CBG plants.

Notes for Facilitation

Start by linking hygiene infrastructure to daily worker behaviour

- Tell participants that hygiene fails first where facilities are missing
- Tell them inspectors assess hygiene within the first few minutes of a site visit
- Explain that minimum infrastructure is a baseline, not a best practice luxury
- List the various risks created when hygiene facilities are absent or poorly maintained

UNIT 5.4.2: Regulatory Expectations (MNRE, CPCB, SPCB, Local Authorities)

Unit Objectives

At the end of this unit, you will be able to:

- Describe the key regulatory authorities governing hygiene and safety in biogas and CBG plants
- Explain the hygiene, safety, and environmental expectations placed on plant operators

Ask

- Ask the participants if regulatory inspections focus only on gas production or also on worker facilities.

Explain

- Explain that biogas and CBG plants operate within a multi-layered regulatory framework.

Notes for Facilitation

- Start by explaining that inspectors judge what they see first

UNIT 5.5: Hazard Identification in Biogas Plants

Unit Objectives

At the end of this unit, you will be able to:

1. Describe why hazard identification is essential in biogas and CBG plant operations
2. Explain how different stages of a biogas plant introduce different types of hazards

Ask

- Ask the participants whether hazards occur only during breakdowns or also during normal plant operation.

Explain

- Explain that a biogas plant is not a single operation but a sequence of interconnected stages.

Notes for Facilitation

- Emphasise that hazard identification is a daily operator responsibility

UNIT 5.5.1: Gas-Related Hazards

Unit Objectives

At the end of this unit, you will be able to:

1. Describe why gas-related hazards are the most critical risks in biogas and CBG plants
2. Explain how failures in gas management lead to serious incidents

Ask

- Ask the participants whether gas hazards occur only during leaks or also during normal operation.

Explain

- Explain that biogas plants are fundamentally gas-centric facilities.

Unit 5.6 Personal Protective Equipment (PPE)

Unit Objectives

At the end of this unit, participants will be able to:

1. Describe the role of PPE as a primary protective barrier in biogas and CBG plants
2. Identify PPE requirements based on different hazard types

Ask

- Ask the participants what PPE they use during their daily plant activities.

Explain

- Explain that Personal Protective Equipment serves as the primary protective barrier between operators and multiple hazards present in biogas and CBG plants.
- Clarify that operators are routinely exposed to toxic gases, corrosive chemicals, wet and slippery surfaces, hot equipment, pressurised systems, and rotating machinery.

Notes for Facilitation

- Emphasise that PPE is mandatory and non-negotiable

Unit 5.6.1 Importance of PPE in Biogas Operations

Unit Objectives

At the end of this unit, participants will be able to:

- Explain why PPE is the last and most critical line of defence in biogas operations
- Identify situations where PPE becomes essential due to residual risk

Ask

- Ask the participants whether all hazards at a biogas plant can be completely eliminated through engineering controls.

Explain

- Explain that biogas O&M teams work in operating zones where hazards cannot be fully eliminated by design or procedures alone.

Notes for Facilitation

- Emphasise PPE as protection against residual risk, not as optional equipment

Unit 5.6.2 Types of PPE and Their Functional Use

Unit Objectives

At the end of this unit, participants will be able to:

- Identify the main categories of PPE used in biogas and CBG plants
- Explain the functional purpose of each PPE category

Ask

- Ask the participants to list the PPE items they normally wear during routine plant work.

Explain

- Explain that PPE in a biogas and CBG plant is grouped into five major categories based on the type of hazard they protect against.
- Clarify that these categories are respiratory protection, hand protection, eye and face protection, body protection, and footwear.

Notes for Facilitation

- Emphasise PPE as a system, not individual equipment
- Use real plant tasks to link hazards with specific PPE
- Encourage participants to check PPE completeness before work
- Clarify that different tasks require different PPE combinations



Figure 16: Components of a PPE kit

Unit 5.6.3 PPE Inspection and Replacement Guidelines

Unit Objectives

At the end of this unit, participants will be able to:

- Explain why PPE inspection at the start of every shift is critical for safe plant operation
- Perform routine inspection checks for commonly used PPE

Ask

- Ask the participants whether PPE inspection is carried out before every shift at their site.

Explain

- Explain that Personal Protective Equipment must be inspected at the start of every shift before entering operational areas.
- Clarify that PPE can degrade due to repeated exposure to gas, chemicals, moisture, heat, and mechanical stress.

Notes for Facilitation

- Emphasise inspection of PPE before every shift without exception.

PPE ITEM	REPLACEMENT FREQUENCY	NOTES
 Respirator cartridges	2–4 weeks or earlier	Replace sooner if odour detected
 Nitrile gloves	After heavy use	Dispose after chemical tasks
 Safety goggles	6–12 months	Replace if scratched
 Safety shoes	1 year	Replace if cracked
 SCBA mask seals	Annually	Must pass leak test
 SCBA cylinders	As per certification	Hydrostatic testing mandatory

Figure 17: PPE Replacement Schedule and Guidelines

Unit 5.6.4 PPE Storage and Hygiene Practices

Unit Objectives

At the end of this unit, participants will be able to:

1. Identify correct storage practices for different PPE types
2. Apply hygiene practices that prevent contamination and material degradation

Ask

- Ask the participants where PPE is stored at their plant
- Explain
- Explain that correct PPE use alone is not sufficient to ensure protection.
- Clarify that incorrect storage or poor hygiene can contaminate PPE and reduce its effectiveness even when it appears intact.
- PPE Storage Requirements

Notes for Facilitation

- Emphasise that PPE care begins after work, not only during use

Team Activity

- Identify proper storage methods for different PPE items.
- Demonstrate correct PPE cleaning and hygiene practices.
- Discuss the impact of poor PPE storage and maintenance on safety.

Unit 5.7 Safe Work Area and Housekeeping Practices

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain why housekeeping is a critical safety and operational control in biogas and CBG plants
2. Identify how poor housekeeping contributes to accidents, equipment failure, and process instability

Ask

- Ask the participants what comes to mind when they hear the term housekeeping.

Explain

- Explain that effective housekeeping in biogas and CBG plants is not merely about cleanliness.

Notes for Facilitation

- Emphasise housekeeping as risk prevention, not cleaning

Unit 5.7.1 Importance of Housekeeping in Biogas O and M

Unit Objectives

At the end of this unit, participants will be able to:

- Explain why housekeeping is a critical safety and operational control in biogas O and M
- Identify common operational risks arising from poor housekeeping

Ask

- Ask the participants if they have experienced slips, blockages, odour issues, or pump problems linked to poor housekeeping.

Explain

- Explain that poor housekeeping is a common root cause of slips, gas accumulation, pest infestation, odour complaints, pump failures, and blocked drains in biogas plants.
- Clarify that when work areas are cluttered or poorly maintained, early warning signs such as leaks, corrosion, or abnormal spills are often overlooked.

Unit 5.7.2 Importance of Safe Work Area

Unit Objectives

At the end of this unit, participants will be able to:

- Explain what defines a safe work area in a biogas or CBG plant
- Identify key physical and visual elements that reduce operational risk

Ask

- Ask the participants what helps them work confidently in a plant area.

Explain

- Explain that a safe work area in a biogas or CBG plant is created through a deliberate combination of cleanliness, orderliness, physical organisation, and clear visual controls.

Notes for Facilitation

- Emphasise that safe work areas are created intentionally through daily discipline



Figure 18: Core Principles of a Safe Work Area

Unit 5.7.3 Housekeeping Practices for Specific Plant Zones

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain why housekeeping requirements differ across plant zones
2. Identify common housekeeping risks associated with specific biogas and CBG plant areas

Ask

- Ask the participants whether all areas of a biogas plant require the same level and type of housekeeping.

Explain

- Explain that housekeeping requirements in biogas and CBG plants vary by zone.
- Clarify that different activities involve different materials, exposure levels, and risk profiles.

Notes for Facilitation

- Emphasise that different zones require different housekeeping focus

Unit 5.8 Gas Leak Detection

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain why gas leak detection is critical in biogas and CBG plant operations
2. Identify the main gases of concern and their associated risks

Ask

- Ask the participants what gases are present in a biogas or CBG plant.

Explain

- Explain that gas leak detection is a critical safety function in biogas and CBG plant operations.
- Clarify that even small and undetected leaks can escalate into serious hazards such as toxic exposure, fire, explosion, or equipment damage.

Notes for Facilitation

- Emphasise that gas leaks are often invisible and odour perception is unreliable

Unit 5.8.1 Understanding Gas Behaviour in Biogas Plants

Unit Objectives

At the end of this unit, participants will be able to:

- Explain how methane and hydrogen sulphide behave when released into air
- Identify areas where different gases are likely to accumulate in a biogas plant

Ask

- Ask the participants whether all gases behave the same way once they leak into the air.

Explain

- Explain that understanding gas behaviour is fundamental to gas safety management in biogas and CBG plants.
- Clarify that methane and hydrogen sulphide behave very differently once released.

Notes for Facilitation

- Emphasise that methane rises and hydrogen sulphide settles

Unit 5.9 Fire Safety and Extinguisher Use

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain why fire safety is critical in biogas and CBG plant operations
2. Identify common fire risk points and ignition sources within the plant

Ask

- Ask the participants what makes biogas plants vulnerable to fire incidents.

Explain

- Explain that fire safety in biogas plants requires strict preventive control because methane, the primary gas produced, is highly flammable.
- Clarify that although biogas plants are designed with ventilation, pressure regulation, and safety systems, fire incidents can still occur.

Notes for Facilitation

- Emphasise that fire prevention starts with gas leak control and good housekeeping
- Use plant layouts to identify fire extinguisher locations
- Highlight that personal safety overrides equipment protection

Unit 5.9.1 Fire Risk Sources in Biogas and CBG Plants

Unit Objectives

At the end of this unit, participants will be able to:

- Identify major fire risk sources in biogas and CBG plants
- Explain how gas, heat, electricity, and combustible materials interact to create fire hazards

Ask

- Ask the participants where they believe the highest fire risk exists in a biogas or CBG plant.

Explain

- Explain that fire risks in biogas and CBG plants arise from the interaction of flammable gases, heat sources, electrical systems, and combustible materials.

Notes for Facilitation

- Emphasise that fires result from combined conditions, not single failures

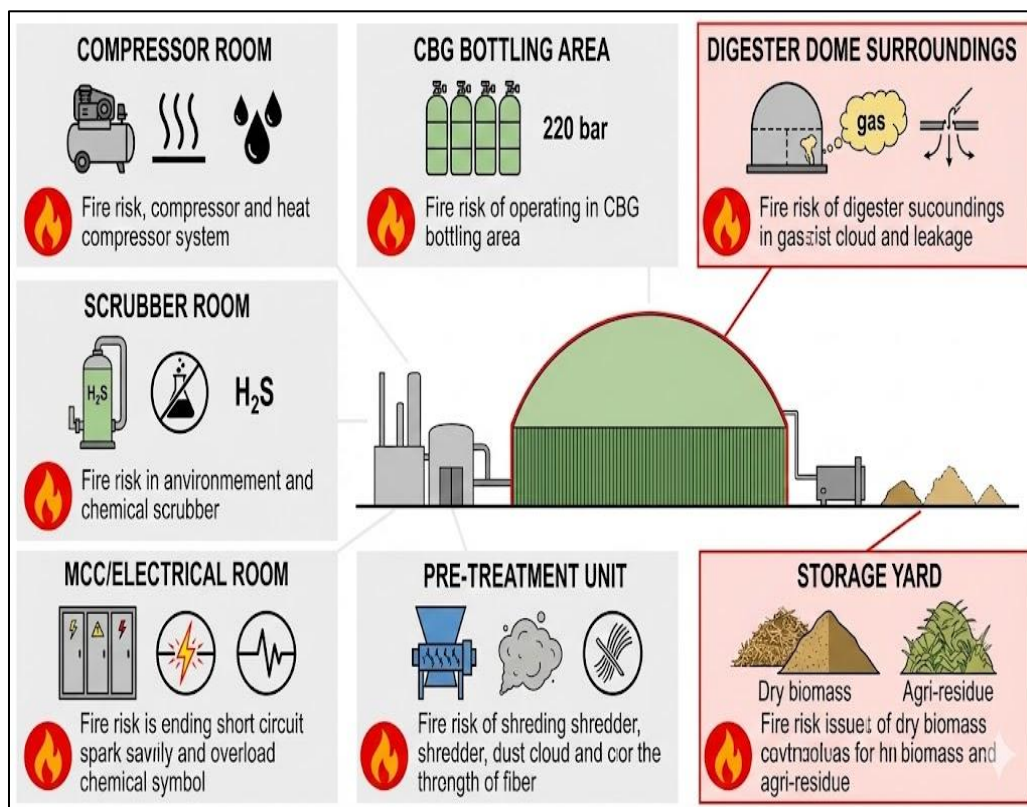


figure 19: High-Risk Fire Zones in or Near Biogas Plant

Unit 5.9.2 Classification of Fire in Biogas Operations

Unit Objectives

At the end of this unit, participants will be able to:

- Explain why fire classification is critical in biogas and CBG plant operations
- Identify different classes of fire commonly encountered in biogas facilities

Ask

- Ask the participants whether all fires can be extinguished using the same method.

Explain

- Explain that different types of fires behave differently depending on the fuel involved.
- Clarify that biogas and CBG plants contain solid biomass, flammable gases, electrical systems, oils, and diesel-based equipment within the same operational space.
- Cause electrical shock

Notes for Facilitation

- Emphasise that correct fire classification prevents escalation

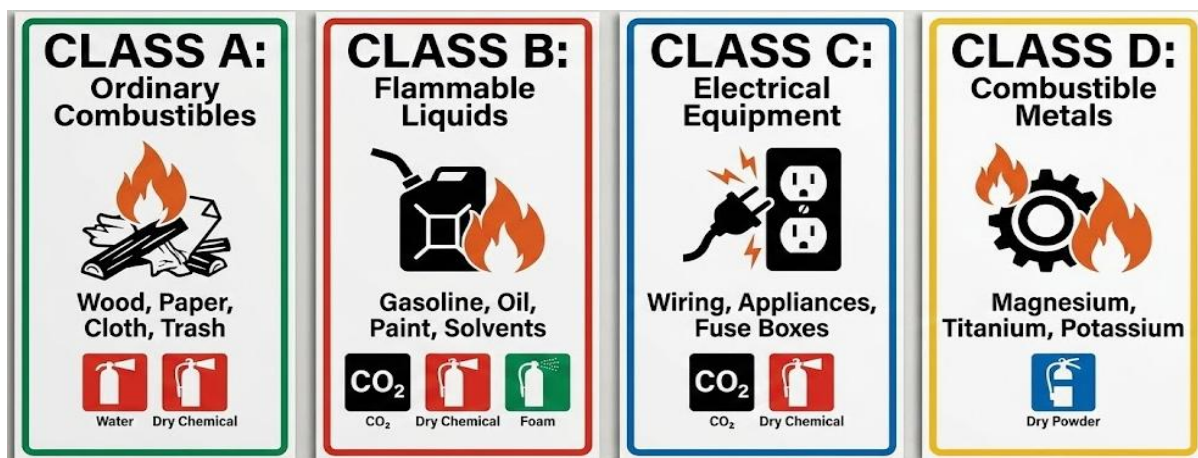


Figure 20: Fire Classes in Biogas Operation

Unit 5.9.3 Types of Fire Extinguishers and Their Applications

Unit Objectives

At the end of this unit, participants will be able to:

- Identify the different types of fire extinguishers used in biogas and CBG plants
- Explain why extinguisher selection must match the fire risk and operating zone

Ask

- Ask the participants whether one fire extinguisher can be safely used for all types of fires.

Explain

- Explain that biogas and CBG plants operate with multiple fire hazards present at the same time.
- Clarify that flammable gases, electrical equipment, liquid fuels, and solid biomass each behave differently when burning.
- The type of fuel involved

Notes for Facilitation

- Emphasize that wrong extinguisher use can increase fire severity

Table 4 : Usage of fire extinguisher

Extinguisher Type	Suitable For	Typical Locations	Why It Is Used
DCP (Dry Chemical Powder – ABC Type)	Methane leaks catching fire, electrical fires, compressor room fires	Compressor room, blower area, digester zone, CHP area	Smothers flames and interrupts chemical reaction; safe for electrical equipment
CO ₂ Extinguisher	Electrical panels, MCC rooms, SCADA rooms	MCC rooms, control panels, analyzer rooms	Leaves no residue, prevents equipment damage
Foam Extinguisher (AFFF)	Oil spills, diesel generator areas	Diesel storage zone, pump room	Spreads a blanket over flammable liquids
Water Extinguisher	Solid fires only (paper, cloth)	General utility areas	Not for electrical or gas fires; used for non-critical zones
			

Unit 5.9.4 Fire Extinguisher Operation (PASS Method)

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain the purpose of the PASS method in fire extinguisher use
2. Demonstrate the correct sequence of extinguisher operation

Ask

- Ask the participants if they have ever used a fire extinguisher during an emergency.

Explain

- Explain that the PASS method is a universally accepted technique for operating fire extinguishers safely and effectively during the initial stage of a fire.
- Clarify that extinguishers work by interrupting combustion at its source.

Notes for Facilitation

- Demonstrate the PASS method using a training extinguisher



Figure 21: PASS Method

Unit 5.9.5 Operational Rules for Fire Safety

Unit Objectives

At the end of this unit, participants will be able to:

1. Explain why disciplined daily practices are critical for fire prevention in biogas and CBG plants
2. Identify operational behaviours that increase or reduce fire risk

Ask

- Ask the participants what daily actions can unintentionally increase fire risk in a plant.

Explain

- Explain that fire safety in biogas and CBG plants depends more on daily discipline than on emergency response alone.
- Clarify that the coexistence of flammable gases, electrical systems, hot equipment, and combustible materials means that small deviations can escalate rapidly.

Notes for Facilitation

- Emphasise that most fires start from ignored warning signs

Safety Area	Operational Rules	Purpose / Risk Controlled
Maintain Safe Work Practices	• Never smoke or use open flames in gas handling or biomass areas • Keep fire exits, pathways, and hydrant points unobstructed • Avoid spark generating metal tools in methane zones • Do not perform welding, cutting, or grinding without a valid hot work permit	Prevents ignition of flammable gases and ensures safe evacuation
Handle Methane Lines and Gas Systems Safely	• Isolate gas valves before any repair or maintenance • Conduct leak testing only with soap solution, never open flames • Drain condensate traps daily to avoid gas accumulation • Stop work immediately if gas smell is detected and investigate safely	Reduces risk of gas buildup, fire, and explosion
Electrical System Safety	• Keep MCC rooms dry, ventilated, and access restricted • Never work on live circuits and follow LOTO procedures • Address loose terminals, overheating, or sparks immediately	Prevents electrical faults and fire initiation





6. SCADA Monitoring, Interpretation & Decision- Making in Biogas Plant O&M

UNIT 6.1 – Overview of SCADA in Biogas and CBG Plant Operations

UNIT 6.2: SCADA Architecture & Components

UNIT 6.3: Real-Time Monitoring Capabilities

Unit 6.4 SCADA Alarms & Alert Management

Unit 6.5 Trend Interpretation & Data Analytics

Unit 6.6-Tiered SCADA Decision-Making Framework

Unit 6.7- SCADA for Predictive & Preventive Maintenance



Key Learning Outcomes



At the end of this module, you will be able to:

1. Identify major hazards in biogas plants, including H₂S, methane leaks, fire risks, and slurry contamination.
2. Use and maintain essential PPE such as masks, gloves, boots, and SCBA sets.
3. Follow hygiene practices required for waste-handling and biogas plant operations.
4. Perform basic gas leak detection and respond to early warning signs.
5. Apply first-aid and emergency procedures for fire, gas exposure, and chemical contact.
6. Execute emergency shutdown steps for leaks, overheating, or high digester pressure.
7. Follow confined-space entry rules and safety reporting protocols.
8. Comply with national safety and environmental guidelines relevant to biogas operations

Key UNIT 6.1 – Overview of SCADA in Biogas and CBG Plant Operations

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Describe why biogas and CBG plants require continuous monitoring due to biological and process variability.
 2. Explain the role of SCADA as a real-time monitoring and control system in biogas and CBG plants.

Ask (Engagement Starter)

- Ask the participants if they know the meaning of SCADA and where they have seen or heard it used before.
- Encourage them to share examples from other industries such as power plants, water treatment plants, or manufacturing units to build familiarity.

Explain (Concept Building)

- Explain that SCADA stands for Supervisory Control and Data Acquisition and is used to continuously monitor, control, and record plant operations.
- Clarify that unlike manual inspections, SCADA works continuously and captures changes that may not be immediately visible to operators on the ground.

Notes for Facilitation

- Start by giving an introduction about how biogas and CBG plants operate under continuously changing conditions due to feedstock variation, microbial activity, and gas handling requirements.
- Tell them that because anaerobic digestion happens inside a sealed digester, operators cannot directly see what is happening inside the system.
- Tell them about the limitations of manual monitoring, such as delayed detection of pH drop, temperature deviation, gas quality deterioration, or equipment stress.

UNIT 6.2: SCADA Architecture & Components

Unit Objectives

- At the end of this unit, you will be able to:
 1. Explain how SCADA captures field data and converts it into monitoring and control actions.
 2. Understand why SCADA systems are organised in layers rather than as a single control unit.

Ask (Participant Engagement)

- When you see values like pH, pressure, or gas flow on the screen, do you know where that information actually comes from?
- Do you think SCADA is only a software screen, or is there more happening behind it?

Explain (Concept Clarification)

- SCADA architecture is not about brand, graphics, or computer screens. It refers to how the entire monitoring and control system is structured, from sensors installed in the field to the decisions taken by controllers and operators. In biogas and CBG plants, where biological and mechanical systems interact continuously, this structure decides how fast problems are detected, how safely equipment is protected, and how reliably the plant operates under changing conditions.

Notes for Facilitation

- Start by giving an introduction about SCADA architecture
- Explain that architecture means the arrangement of components and information flow. Emphasise that SCADA architecture defines how data is captured, processed, displayed, and acted upon.
- Tell them that biogas and CBG plants operate under continuously changing conditions

UNIT 6.3: Real-Time Monitoring Capabilities

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Describe what real-time monitoring means in the context of SCADA-operated biogas and CBG plants.
 2. Explain why continuous, second-by-second visibility is critical for biological stability and equipment protection.

Ask (Participant Engagement)

- How often do you currently check pH, temperature, gas pressure, or compressor status manually?
- What happens if a parameter changes at night or between shifts?

Explain (Concept Clarification)

- Real-time monitoring is the foundation on which all SCADA functions are built. Without live, continuous data, automation, alarms, and analytics lose their effectiveness. In biogas and CBG plants, where biological and mechanical systems react slowly at first but fail rapidly once limits are crossed, early visibility is the difference between control and crisis.

Notes for Facilitation

- Start by explaining the limitations of manual and periodic monitoring
- Tell them that anaerobic digestion does not fail suddenly, it drifts first

Unit 6.4 SCADA Alarms & Alert Management

Unit Objectives

- At the end of this unit, participants will be able to:
 1. Describe the purpose and importance of SCADA alarms in biogas and CBG plant operation.
 2. Explain how alarms act as an early-warning system for biological, mechanical, and safety risks.
 3. Differentiate between alarm types, priorities, and operator response expectations.

Notes for Facilitation

- Start by explaining that alarms are decision triggers, not noise
- Tell them that every alarm has a reason and a consequence
- Tell them that delayed response turns minor deviations into major failures
- Tell them that acknowledging an alarm is not the same as resolving it
- Tell them that disciplined alarm handling protects both biology and machines
- Explain that good alarm management reduces stress and confusion during abnormal conditions

Facilitator Closing Emphasis

- SCADA alarms are the plant's early-warning intelligence system. Operators who respect alarms, understand their meaning, and respond correctly prevent accidents, protect microbes, extend equipment life, and keep the plant stable. Poor alarm management does not fail immediately, but it always fails eventually. Strong alarm discipline is a hallmark of a safe, reliable, and well-run biogas or CBG plant.

Unit 6.5 Trend Interpretation & Data Analytics

Unit Objectives

1. Describe what trend interpretation means in the context of SCADA-based biogas and CBG plant operation.
2. Explain how gradual changes in process trends act as early warning signals for biological, mechanical, and gas quality issues.

Ask

- Ask the participants if they have ever noticed slow changes in parameters such as temperature, pH, gas flow, or H₂S that later resulted in a major issue.
- Ask them whether they usually focus only on current values or also look at trends over hours, days, or weeks.

Explain

- Explain that trend interpretation is the skill of reading how parameters change over time, not just what their current value is. In biogas plants, most problems do not appear suddenly. The digester gives early signals through slow drifts and repeated patterns. SCADA trends make these signals visible, allowing operators to act early instead of waiting for alarms or breakdowns.

Notes for Facilitation

- Start by giving an introduction about why anaerobic digestion is a slow, biologically driven process where failures usually develop gradually rather than instantly.
- Tell them that raw SCADA numbers by themselves do not explain plant behaviour, but trends reveal whether the process is stable, improving, or slowly moving toward trouble.
- Tell them about the typical early warning trends seen in biogas plants, such as gradual temperature decline, slow pH drift, increasing H₂S levels, fluctuating gas flow, or rising compressor current.

Unit 6.6-Tiered SCADA Decision-Making Framework

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe the concept of tiered decision-making in SCADA-based biogas and CBG plant operations.
 2. Explain why different levels of deviations require different levels of operator response.
 3. Differentiate between Basic, Intermediate, and Advanced decision tiers in daily plant operation.
 4. Apply the correct decision level for alarms, trends, and deviations without overreacting or delaying action.

Ask

- Ask the participants if all SCADA alarms and deviations should be handled in the same way by every operator.
- Ask them if they have seen situations where either too much action or too little action was taken because roles were unclear.

Explain

- Explain that SCADA does not replace human decision-making but structures it. In a biogas or CBG plant, not every deviation needs escalation, and not every operator should take the same type of action. The tiered decision-making framework ensures that responses are proportional to risk, aligned with competence, and consistent across shifts. This prevents both panic-driven overcorrection and dangerous delays.

Notes for Facilitation

- Start by giving an introduction about why unstructured decision-making is risky in biogas plants.
- Explain that relying only on personal experience leads to inconsistent responses, shift-to-shift variation, and avoidable process shocks.
- Tell them that the tiered framework divides decisions into three clear levels: Basic, Intermediate, and Advanced.

6.7 SCADA for Predictive & Preventive Maintenance

Unit Objectives

- At the end of this unit, you will be able to:
 1. Describe how SCADA supports predictive and preventive maintenance in biogas and CBG plants.
 2. Explain the difference between reactive, preventive and predictive maintenance using SCADA data.

Ask

- Ask the participants if most equipment failures happen suddenly or develop gradually.
- Ask them how they currently decide when to service compressors, pumps or agitators.

Explain

- Explain that SCADA enables maintenance teams to see how equipment is ageing while it is still running safely. Instead of waiting for breakdowns or following fixed calendar schedules, SCADA allows maintenance to be planned based on real operating behaviour, trends and early deviation signals.

Notes for Facilitation

- Start by giving an introduction about maintenance challenges in biogas and CBG plants. Biological variability, moisture, corrosive gases and continuous operation place high stress on equipment, making early detection essential.
- Tell them that SCADA shifts maintenance from reactive repair to predictive and preventive action.
Breakdowns are costly, disruptive and often linked to safety risks.

Facilitator Closing Note

- Emphasise that predictive and preventive maintenance is not about adding more work. It is about doing the right work at the right time, guided by SCADA data. Plants that use SCADA intelligently for maintenance move from firefighting to foresight, achieving safer, more reliable and cost-effective biogas and CBG operations.





Annexure



CHAPTER	UNIT NAME	HEADING	YOUTUBE LINK	QR CODE
CHAPTER 1		Basics of Biogas Technology	https://www.youtube.com/watch?v=ZQNkV-u8Rq0	
CHAPTER 1	UNIT – 1.5	Composition, Properties, and Energy Value of Biogas	https://www.youtube.com/watch?v=pB3jZB-9TXA	
CHAPTER -2	UNIT – 2.1	feedstock management and pre-treatment in biogas plant	https://www.youtube.com/watch?v=wlsTOD0wXNI	
CHAPTER -2	UNIT – 2.6	Pre-treatment for Feedstock	https://www.youtube.com/watch?v=5FqUxeWhJZ0	
CHAPTEER -3	UNIT -3.1	FIXED DOME DIGESTER	https://www.youtube.com/watch?v=HyJOtrov2g8	
CHAPTER – 3	UNIT -3.2	FLOATING DOME	https://www.youtube.com/watch?v=OC4OArm6da8	
CHAPTER 3	UNIT -3.3	CSTR Biogas Digesters	https://www.youtube.com/watch?v=yWW0fsYI4NA	

CHAPTER -4	UNIT -3.4	Plug-Flow Biogas Digester	https://www.youtube.com/watch?v=2AqhQmalMug	
CHAPTER -5	UNIT -3.5	CHP - Combined Heat and Power	https://www.youtube.com/watch?v=2Kc6xKQIDtU&t=40s	
CHAPTER -4	UNIT 4.1	Monitoring of Biogas Plants	https://www.youtube.com/watch?v=mxjSPu8e1gl	
CHAPTER -4	UNIT -4.8	Seasonal O&M Adjustments OF BIOGAS	https://www.youtube.com/watch?v=XCUvowi2ZIU	
CHAPTER – 5	UNIT 5.1	Introduction to Health and Hygiene Fundamentals in Biogas Environments	https://www.youtube.com/watch?v=LhXOY9O0Ulw	
CHAPTER -6	UNIT 6.1	SCADA IN BIOGAS	https://www.youtube.com/watch?v=HeDTKwffN-s&t=85s	

CHAPTER 6	UNIT -6.4	SCADA Alarms & Alert Management	https://www.youtube.com/watch?v=tep95WjpBIU	
-----------	-----------	---------------------------------	---	---



Skill India

कौशल भारत - कुशल भारत



Please Scan/ Click QR Code to access e book



Address: 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri New Delhi - 110021

Email: info@sscgj.in

Web: www.sscgj.in

Phone: 011-41792866

CIN No: 00000000

Price : ₹